```
# ==== run.py ====
# -*- coding: utf-8 -*-
"""系统启动入口。
运行：
    python run.py                启动并自动打开浏览器
    python run.py --no-browser   只启动服务，不打开浏览器
    python run.py --dev          开发模式（开启代码热重载与调试器）
    python run.py --port 5001    指定端口
默认以 127.0.0.1:5000 启动，就绪后自动打开浏览器。
"""
import os
import sys
import threading
import time
import webbrowser
if hasattr(sys.stdout, "reconfigure"):
    sys.stdout.reconfigure(encoding="utf-8")   # 控制台中文正常显示
from app import create_app
from app.models import db
PORT = 5000
for i, a in enumerate(sys.argv):
    if a == "--port" and i + 1 < len(sys.argv):
        PORT = int(sys.argv[i + 1])
URL = "http://127.0.0.1:%d" % PORT
NO_BROWSER = "--no-browser" in sys.argv
DEV = "--dev" in sys.argv
app = create_app()
def open_browser_later(delay=1.2):
    """等服务工作起来再打开浏览器，避免打开时页面还没就绪。
    放在后台线程里，不用 sleep 阻塞服务启动；线程设为守护线程，
    用户 Ctrl+C 停止服务时不会残留。
    """
    def run():
        time.sleep(delay)
        try:
            webbrowser.open(URL)
        except Exception:
            pass            # 打开失败不影响服务本身
    t = threading.Thread(target=run, daemon=True)
    t.start()
if __name__ == "__main__":
    with app.app_context():
        db.create_all()     # 首次运行时自动建表
        from app.utils.schema import ensure_schema
        added = ensure_schema(db)     # 旧库补齐新增字段
        if added:
            print("已补充字段：%s" % "、".join(added))
    print("=" * 58)
    print("   历史遗留废弃矿山自然恢复适宜性评价系统  Ｖ１．０")
    print("=" * 58)
```

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    print("   访问地址：%s" % URL)
    print("   演示账号：admin      密码：admin123")
    print("   停止运行：按 Ctrl+C，或直接关闭本窗口")
    print("=" * 58)
    if not NO_BROWSER and not DEV:
        print("   正在打开浏览器……")
    print()
    # 开发模式才启用热重载（重载会二次启动进程，故此时不自动开浏览器）
    if not NO_BROWSER and not DEV:
        open_browser_later()
    app.run(host="127.0.0.1", port=PORT, debug=DEV,
    use_reloader=DEV)
# ==== app/__init__.py ====
# -*- coding: utf-8 -*-
"""Flask 应用工厂。"""
from datetime import date
from flask import Flask
from flask_login import LoginManager
from app.config import Config
from app.models import db, User
from app.utils.constants import ROLE_LABELS
login_manager = LoginManager()
login_manager.login_view = "auth.login"
login_manager.login_message = "请先登录"
@login_manager.user_loader
def load_user(user_id):
    return db.session.get(User, int(user_id))
def create_app(config_class=Config):
    app = Flask(__name__)
    app.config.from_object(config_class)
    db.init_app(app)
    login_manager.init_app(app)
    # 注册视图（各模块路由）
    from app.views import auth, project, unit, standard,
    constraint, \
        ahp, indicator, evaluate, report, search
    app.register_blueprint(auth.bp)
    app.register_blueprint(project.bp)
    app.register_blueprint(unit.bp)
    app.register_blueprint(standard.bp)
    app.register_blueprint(constraint.bp)
    app.register_blueprint(ahp.bp)
    app.register_blueprint(indicator.bp)
    app.register_blueprint(evaluate.bp)
    app.register_blueprint(report.bp)
    app.register_blueprint(search.bp)
    # 全局登录保护：除登录页与静态资源外，均需登录
    from flask import request, redirect, url_for,
    send_from_directory
    from flask_login import current_user, login_user
```

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PUBLIC_ENDPOINTS = {"auth.login", "auth.register",
"static", "favicon"}
# 浏览器默认会请求根路径下的 /favicon.ico。
# 页面里虽已用 <link rel="icon"> 指向 static 目录，但出错页等场景仍会
# 回退到这个路径，补一条路由免得日志里一直刷 404。
@app.route("/favicon.ico")
def favicon():
    return send_from_directory(app.static_folder,
"favicon.ico",
                               mimetype="image/vnd.micro
soft.icon")
@app.before_request
def require_login():
    # 截图模式（仅用于生成说明书配图）：自动以管理员身份登录
    if app.config.get("SCREENSHOT_MODE"):
        if (request.endpoint not in ("auth.login",
"auth.register", "static")
                and not current_user.is_authenticated):
            admin = User.query.filter_by(role="admin").f
irst()
            if admin:
                login_user(admin)
        return None
    if request.endpoint in PUBLIC_ENDPOINTS or
request.endpoint is None:
        return None
    if not current_user.is_authenticated:
        return redirect(url_for("auth.login"))
    return None
# 侧边栏导航：按论文评价流程的顺序编排，分为「评价流程」与「系统管理」两组。
# 不使用模块编号，避免出现与实际流程不一致的代号。
# 每项带图标名（内联 SVG，见 templates/_macros.html 的 ICONS）
# 与可选的待办角标（该步骤还没做完的图斑数）。
FLOW_LINKS = [
    ("项目与矿山档案", "/project/", "folder"),
    ("政策与标准知识库", "/standard/", "book"),
    ("限制性指标判定", "/constraint/", "shield"),
    ("微地质单元", "/unit/", "layers"),
    ("指标体系与权重", "/ahp/", "sliders"),
    ("指标数据采集", "/indicator/", "database"),
    ("适宜性定量评价", "/evaluate/", "target"),
    ("报告与结果输出", "/report/", "file"),
]
# 「用户与操作日志」已并入「个人设置」，不再是独立菜单，
# 因此系统管理组只留一个入口。
ADMIN_LINKS = [
    ("个人设置", "/profile", "user"),
]
@app.context_processor
def inject_nav():
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    path = request.path or "/"
    def hit(prefix):
        """当前路径是否属于某个入口（/unit/ 也匹配 /unit/3）。"""
        return path.startswith(prefix.rstrip("/"))
    badges = {}
    if current_user.is_authenticated:
        from app.services import dashboard as dash
        badges = dash.nav_badges()
    def build(links):
        return [{"label": label, "href": href, "icon": icon,
                 "active": hit(href), "badge": badges.get(href, 0)}
                for label, href, icon in links]
    groups = [
        {"title": "评价流程", "links": build(FLOW_LINKS)},
        {"title": "系统管理", "links": build(ADMIN_LINKS)},
    ]
    # 面包屑用的当前页名：按路径命中的入口取，未命中时给个通用名
    if path == "/":
        page_title = "工作台"
    elif path.startswith("/profile"):
        page_title = "个人设置"
    elif path.startswith(("/users", "/logs")):
        page_title = "账号与日志"
    elif path.startswith("/search"):
        page_title = "搜索结果"
    else:
        page_title = "工作台"
        for label, href, _icon in FLOW_LINKS:
            if hit(href):
                page_title = label
                break
    # 层级引导用：项目是归档单位，图斑才是评价对象。
    # 下游各页面按图斑组织数据，因此需要提示「哪些项目下还没有图斑」。
    from app.models import Project, MinePatch
    projects = Project.query.order_by(Project.id).all()
    return {
        "nav_groups": groups,
        "home_active": path == "/",
        "page_title": page_title,
        "today": date.today().strftime("%Y-%m-%d"),
        "search_q": request.args.get("q", ""),
        "role_labels": ROLE_LABELS,
        "project_count": len(projects),
        "patch_count": MinePatch.query.count(),
        "projects_without_patch": [p for p in projects
                                   if p.patches.count() == 0],
    }
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    # 操作审计：统一在响应阶段记录成功的写操作（操作日志）
    from app.services import audit
    @app.after_request
    def audit_writes(response):
        if request.method in ("POST", "PUT", "PATCH",
    "DELETE"):
            audit.record(request.endpoint,
    response.status_code)
        return response
    # 错误页：给出可读提示，而不是默认的英文错误页
    from flask import render_template
    @app.errorhandler(403)
    def forbidden(e):
        return render_template("error.html", code=403,
                               message="没有权限执行该操作"), 403
    @app.errorhandler(404)
    def not_found(e):
        return render_template("error.html", code=404,
                               message="请求的页面不存在"), 404
    return app
# ==== app/config.py ====
# -*- coding: utf-8 -*-
"""应用配置。"""
import os
from datetime import timedelta
BASE_DIR = os.path.abspath(os.path.dirname(__file__))
class Config:
    SECRET_KEY = os.environ.get("SECRET_KEY",
    "mine-restoration-dev-key")
    # 单机部署，SQLite 数据库文件放在应用目录下
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE_DIR, "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    # 禁止跨站请求伪造（CSRF），登录与表单提交依赖此项
    WTF_CSRF_ENABLED = True
    # 「记住我」：登录时勾选则下发持久 cookie，关闭浏览器后仍可自动登录
    REMEMBER_COOKIE_DURATION = timedelta(days=30)
    REMEMBER_COOKIE_HTTPONLY = True
    REMEMBER_COOKIE_SAMESITE = "Lax"
    # 影像与数据文件上传（遥感影像挂接、批量导入）
    UPLOAD_FOLDER = os.path.join(BASE_DIR, "static",
    "uploads")
    MAX_CONTENT_LENGTH = 32 * 1024 * 1024          # 单次上传上限
    32 MB
    ALLOWED_IMAGE_EXT = {"png", "jpg", "jpeg", "gif",
    "webp", "bmp", "tif", "tiff"}
    ALLOWED_DATA_EXT = {"csv", "xlsx", "xls"}
    # 头像只用浏览器能直接显示的格式（不含 bmp / tif），并单独限制体积
    ALLOWED_AVATAR_EXT = {"png", "jpg", "jpeg", "gif",
    "webp"}
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    MAX_AVATAR_LENGTH = 4 * 1024 * 1024          # 4 MB
# ==== app/models.py ====
# -*- coding: utf-8 -*-
"""数据模型：系统全部数据表。
本文件由甲在阶段 0 一次性写完，作为两人独立开发的地基。
约定：两人只读此文件；确需改表时，由乙提需求、甲统一修改。
"""
import json
from datetime import datetime
from flask_login import UserMixin
from flask_sqlalchemy import SQLAlchemy
db = SQLAlchemy()
class User(UserMixin, db.Model):
    """用户（账户与权限）。"""
    __tablename__ = "users"
    id = db.Column(db.Integer, primary_key=True)
    username = db.Column(db.String(64), unique=True,
    nullable=False, index=True)
    password_hash = db.Column(db.String(256),
    nullable=False)
    # 职务只有两种：管理员 admin（可管理账号与他人的职务） / 操作员 operator
    role = db.Column(db.String(20), nullable=False,
    default="operator")
    active = db.Column(db.Boolean, nullable=False,
    default=True)
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
    # ---- 个人资料（用户可在「个人设置」里自行维护）----
    display_name = db.Column(db.String(64), nullable=True)
    # 昵称
    email = db.Column(db.String(128), nullable=True)
    phone = db.Column(db.String(32), nullable=True)
    gender = db.Column(db.String(8), nullable=True)
    # 男 / 女 / 保密
    avatar = db.Column(db.String(256), nullable=True)
    # 头像，相对 static 的路径
    @property
    def is_admin(self):
        return self.role == "admin"
    @property
    def shown_name(self):
        """界面上显示的名字：优先用昵称，没填就用账号名。"""
        return self.display_name or self.username
    @property
    def initial(self):
        """没有头像时用首字占位。"""
        return (self.shown_name or "?")[0].upper()
    def to_dict(self):
        return {"id": self.id, "username": self.username,
                "display_name": self.display_name,
```

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        "gender": self.gender,                                          301
                    "role": self.role, "active": self.active}           302
class OperationLog(db.Model):                                           303
    """操作日志。"""                                                    304
    __tablename__ = "operation_logs"                                    305
    id = db.Column(db.Integer, primary_key=True)                        306
    user_id = db.Column(db.Integer, db.ForeignKey("users.id"            307
    ), nullable=True)                                                   308
    action = db.Column(db.String(128), nullable=False)                  309
    # 操作类型                                                          310
    target_type = db.Column(db.String(64), nullable=True)               311
    # 对象类型                                                          312
    target_id = db.Column(db.Integer, nullable=True)                    313
    # 对象主键                                                          314
    detail = db.Column(db.Text, nullable=True)                          315
    # 操作详情                                                          316
    created_at = db.Column(db.DateTime,                                 317
    default=datetime.utcnow, index=True)                                318
class Project(db.Model):                                                319
    """评价项目（项目与矿山档案）。"""                                  320
    __tablename__ = "projects"                                          321
    id = db.Column(db.Integer, primary_key=True)                        322
    name = db.Column(db.String(128), nullable=False)                    323
    code = db.Column(db.String(64), nullable=True)                      324
    # 项目编号                                                          325
    description = db.Column(db.Text, nullable=True)                     326
    status = db.Column(db.String(20), default="进行中")                 327
    # 进行中 / 已完成 / 已归档                                          328
    created_by = db.Column(db.Integer,                                  329
    db.ForeignKey("users.id"), nullable=True)                           330
    created_at = db.Column(db.DateTime,                                 331
    default=datetime.utcnow)                                            332
    patches = db.relationship("MinePatch",                              333
    backref="project", lazy="dynamic",                                  334
                                    cascade="all, delete-orphan")       335
class MinePatch(db.Model):                                              336
    """矿山图斑。"""                                                    337
    __tablename__ = "mine_patches"                                      338
    id = db.Column(db.Integer, primary_key=True)                        339
    project_id = db.Column(db.Integer,                                  340
    db.ForeignKey("projects.id"), nullable=False)                       341
    name = db.Column(db.String(128), nullable=False)                    342
    admin_region = db.Column(db.String(128), nullable=True)             343
    # 所在行政区                                                        344
    center_x = db.Column(db.Float, nullable=True)                       345
    # 中心经度 / 横坐标                                                 346
    center_y = db.Column(db.Float, nullable=True)                       347
    # 中心纬度 / 纵坐标                                                 348
    area_ha = db.Column(db.Float, nullable=True)                        349
    # 图斑面积（公顷）                                                  350
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    mineral_type = db.Column(db.String(64), nullable=True)
    # 开采矿种
    mining_method = db.Column(db.String(64), nullable=True)
    # 开采方式
    stop_years = db.Column(db.Float, nullable=True)
    # 停产年限
    responsible_entity_missing = db.Column(db.Boolean,
    default=True)  # 责任主体是否缺失
    # 遥感底图 / 无人机正射影像挂接
    image_path = db.Column(db.String(256), nullable=True)
    # 相对 static 的路径
    image_note = db.Column(db.String(256), nullable=True)
    # 影像说明（如来源、时相）
    image_uploaded_at = db.Column(db.DateTime,
    nullable=True)
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
    units = db.relationship("MicroUnit", backref="patch",
    lazy="dynamic",
                            cascade="all, delete-orphan")
class ReferenceEcosystem(db.Model):
    """参照生态系统：标准化基准值，与图斑绑定。"""
    __tablename__ = "reference_ecosystems"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    coverage_benchmark = db.Column(db.Float, default=75.0)
    # 植被覆盖度基准 %
    shannon_benchmark = db.Column(db.Float, default=0.7)
    # Shannon-Wiener 基准
    dominant_species = db.Column(db.String(256),
    nullable=True)  # 建群种
    soil_benchmark = db.Column(db.String(256),
    nullable=True)    # 土壤基准
class Standard(db.Model):
    """地方标准 / 政策文件（政策与标准知识库）。"""
    __tablename__ = "standards"
    id = db.Column(db.Integer, primary_key=True)
    province = db.Column(db.String(32), nullable=False)
    # 省份
    name = db.Column(db.String(256), nullable=False)
    # 名称
    doc_number = db.Column(db.String(128), nullable=True)
    # 标准编号 / 文号
    publish_date = db.Column(db.String(32), nullable=True)
    implement_date = db.Column(db.String(32), nullable=True)
    # 类别：地方标准 / 政策文件
    category = db.Column(db.String(32), nullable=False,
    default="地方标准")
    summary = db.Column(db.Text, nullable=True)
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    # 特点概述
    rules = db.relationship("StandardRule",
    backref="standard",
                            cascade="all, delete-orphan")
class StandardRule(db.Model):
    """结构化规则：将标准条文拆解为可执行规则。"""
    __tablename__ = "standard_rules"
    id = db.Column(db.Integer, primary_key=True)
    standard_id = db.Column(db.Integer,
    db.ForeignKey("standards.id"), nullable=False)
    item = db.Column(db.String(128), nullable=False)
    # 判定项
    operator = db.Column(db.String(8), nullable=False)
    # 比较符：> >= < <= ==
    threshold = db.Column(db.String(64), nullable=True)
    # 阈值
    condition_type = db.Column(db.String(32),
    default="必要条件")  # 必要条件 / 充分条件
    region = db.Column(db.String(64), nullable=True)
    # 适用分区（宁夏分区差异化）
    note = db.Column(db.Text, nullable=True)
class ConstraintCheck(db.Model):
    """限制性判定记录。"""
    __tablename__ = "constraint_checks"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    policy_pass = db.Column(db.Boolean, nullable=True)
    # 政策约束
    location_pass = db.Column(db.Boolean, nullable=True)
    # 区位约束
    pollution_pass = db.Column(db.Boolean, nullable=True)
    # 土壤污染
    geo_pass = db.Column(db.Boolean, nullable=True)
    # 地质灾害
    farmland_pass = db.Column(db.Boolean, nullable=True)
    # 耕地属性
    passed = db.Column(db.Boolean, nullable=True)
    # 总体是否通过
    reason = db.Column(db.Text, nullable=True)
    # 不通过原因
    checked_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class MicroUnit(db.Model):
    """微地质单元：岩质边坡 / 碎石边坡 / 平台。"""
    __tablename__ = "micro_units"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    unit_type = db.Column(db.String(32), nullable=False)
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    # 单元类型
    area = db.Column(db.Float, nullable=True)
    # 面积
    ratio = db.Column(db.Float, nullable=True)
    # 面积占比（自动计算）
    description = db.Column(db.Text, nullable=True)
    # 现场描述
class Indicator(db.Model):
    """指标定义：指标树，含角色与启用单元类型。"""
    __tablename__ = "indicators"
    id = db.Column(db.Integer, primary_key=True)
    code = db.Column(db.String(32), unique=True,
    nullable=False)  # 指标编码
    name = db.Column(db.String(64), nullable=False)
    # 指标名称
    level = db.Column(db.Integer, nullable=False)
    # 1/2/3 级
    parent_code = db.Column(db.String(32), nullable=True)
    # 父指标编码
    # 角色：土壤污染指标同时作为限制性指标与评价指标
    is_constraint = db.Column(db.Boolean, default=False)
    is_evaluative = db.Column(db.Boolean, default=True)
    unit = db.Column(db.String(32), nullable=True)
    # 单位
    ideal_value = db.Column(db.String(64), nullable=True)
    # 理想值（可配置）
    ideal_type = db.Column(db.String(32), nullable=True)
    # benefit/cost/mid/discrete
    # 启用该指标的微地质单元（逗号分隔）。岩质边坡不启用土壤指标、平台不启用地形指标
    enabled_units = db.Column(db.String(128), nullable=True)
    description = db.Column(db.Text, nullable=True)
    # 取值依据说明
    def units_list(self):
        return [s for s in (self.enabled_units or
    "").split(",") if s]
class WeightScheme(db.Model):
    """权重方案：按微地质单元类型分别保存。"""
    __tablename__ = "weight_schemes"
    id = db.Column(db.Integer, primary_key=True)
    unit_type = db.Column(db.String(32), nullable=False)
    # 所属单元类型
    name = db.Column(db.String(64), nullable=True)
    # 方案名称
    level = db.Column(db.Integer, nullable=False,
    default=1)        # 1=一级指标 2=二级指标
    is_default = db.Column(db.Boolean, default=False)
    # 是否默认（论文表 3）
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class WeightItem(db.Model):
```

```
    """权重明细：方案内的各项权重。"""                                    501
    __tablename__ = "weight_items"                                      502
    id = db.Column(db.Integer, primary_key=True)                        503
    scheme_id = db.Column(db.Integer,                                   504
    db.ForeignKey("weight_schemes.id"), nullable=False)                 505
    indicator_code = db.Column(db.String(32),                           506
    nullable=False)       # 指标编码                                    507
    weight = db.Column(db.Float, nullable=False)                        508
    # 权重值                                                            509
class ExpertMatrix(db.Model):                                           510
    """专家判断矩阵：多专家打分的原始矩阵。"""                            511
    __tablename__ = "expert_matrices"                                   512
    id = db.Column(db.Integer, primary_key=True)                        513
    scheme_id = db.Column(db.Integer,                                   514
    db.ForeignKey("weight_schemes.id"), nullable=True)                  515
    # 直接标注适用对象，便于无需先建权重方案即可录入专家矩阵             516
    unit_type = db.Column(db.String(32), nullable=True)                 517
    # 微地质单元类型                                                    518
    l1_name = db.Column(db.String(64), nullable=True)                   519
    # 所属一级指标（二级矩阵用）                                        520
    expert_name = db.Column(db.String(64), nullable=True)               521
    # 专家姓名                                                          522
    level = db.Column(db.Integer, nullable=False,                       523
    default=1)          # 矩阵对应指标层级                              524
    matrix_json = db.Column(db.Text, nullable=False)                    525
    # 判断矩阵（JSON 二维数组）                                         526
    cr = db.Column(db.Float, nullable=True)                             527
    # 该专家矩阵的一致性比例                                            528
    created_at = db.Column(db.DateTime,                                 529
    default=datetime.utcnow)                                            530
class IndicatorData(db.Model):                                          531
    """指标实测数据：按微地质单元组织。"""                               532
    __tablename__ = "indicator_data"                                    533
    id = db.Column(db.Integer, primary_key=True)                        534
    unit_id = db.Column(db.Integer, db.ForeignKey("micro_uni            535
    ts.id"), nullable=False, unique=True)                               536
    soil_ratio = db.Column(db.Float, nullable=True)                     537
    # 土石比                                                            538
    thickness = db.Column(db.Float, nullable=True)                      539
    # 土层厚度 cm                                                       540
    ph = db.Column(db.Float, nullable=True)                             541
    organic_matter = db.Column(db.Float, nullable=True)                 542
    # 有机质 g/kg                                                       543
    # 重金属含量，JSON 字符串：{"Hg": 0.05, ...}，未检测项缺失或为 null   544
    heavy_metals = db.Column(db.Text, nullable=True)                    545
    coverage = db.Column(db.Float, nullable=True)                       546
    # 植被覆盖度 %                                                      547
    shannon = db.Column(db.Float, nullable=True)                        548
    # Shannon-Wiener 多样性指数                                         549
    slope = db.Column(db.Float, nullable=True)                          550
```

```
    # 坡度 °
    aspect = db.Column(db.String(16), nullable=True)
    # 坡向：阳坡 / 阴阳坡 / 阴坡
    def heavy_metals_dict(self):
        return json.loads(self.heavy_metals) if
    self.heavy_metals else {}
class EvaluationResult(db.Model):
    """评价结果。"""
    __tablename__ = "evaluation_results"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    final_score = db.Column(db.Float, nullable=True)
    # 最终评价因子 F
    grade = db.Column(db.String(32), nullable=True)
    # 适宜性等级
    measures = db.Column(db.Text, nullable=True)
    # 推荐措施
    unit_scores = db.Column(db.Text, nullable=True)
    # 各单元得分 JSON
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class ScoreDetail(db.Model):
    """得分明细：每一层指标的权重、标准化值、加权贡献，供追溯与复核。"""
    __tablename__ = "score_details"
    id = db.Column(db.Integer, primary_key=True)
    result_id = db.Column(db.Integer,
    db.ForeignKey("evaluation_results.id"), nullable=False)
    unit_id = db.Column(db.Integer, nullable=True)
    l1_name = db.Column(db.String(64), nullable=True)
    # 一级指标
    l2_name = db.Column(db.String(64), nullable=True)
    # 二级指标
    raw_value = db.Column(db.String(64), nullable=True)
    # 实测值
    std_value = db.Column(db.Float, nullable=True)
    # 标准化值
    weight = db.Column(db.Float, nullable=True)
    # 权重
    contribution = db.Column(db.Float, nullable=True)
    # 加权贡献
# ==== app/forms.py ====
# -*- coding: utf-8 -*-
"""表单定义（共享，由甲统一维护）。
各模块的 Flask-WTF 表单集中于此。
"""
import re
from flask_wtf import FlaskForm
from wtforms import SelectField, SubmitField, StringField,
    PasswordField, BooleanField
```

```
from wtforms.validators import (DataRequired, Length,
    Optional, EqualTo,
                                ValidationError)
from app.utils.constants import UNIT_TYPES, ASPECT_TYPES,
    GENDERS
#  邮箱格式：够用的轻量校验。
#  WTForms 自带的 Email() 需要额外安装 email_validator，
#  为一个字段引入依赖不划算，这里自己用正则判断。
EMAIL_RE = re.compile(r"^[^@\s]+@[^@\s]+\.[A-Za-z]{2,}$")
def email_format(form, field):
    v = (field.data or "").strip()
    if v and not EMAIL_RE.match(v):
        raise ValidationError("邮箱格式不正确")
def email_ok(value):
    """邮箱格式判断，供不用 WTForms 的表单复用（如管理员编辑用户）。"""
    v = (value or "").strip()
    return (not v) or bool(EMAIL_RE.match(v))
class AHPSelectForm(FlaskForm):
    """选择微地质单元类型与指标层级，以确定判断矩阵规模。"""
    unit_type = SelectField("微地质单元类型",
    validators=[DataRequired()])
    level = SelectField("指标层级",
                        choices=[("1", "一级指标"), ("2",
    "二级指标")],
                        validators=[DataRequired()])
    l1 = SelectField("一级指标", validators=[],
    validate_choice=False)  # 动态填充，层级=2 时使用
    submit = SubmitField("生成判断矩阵")
    def __init__(self, *args, **kwargs):
        super().__init__(*args, **kwargs)
        self.unit_type.choices = [(t, t) for t in
    UNIT_TYPES]
class EmptyForm(FlaskForm):
    """仅用于承载 CSRF 令牌的空表单（矩阵录入等自定义表单使用）。"""
    pass
class LoginForm(FlaskForm):
    """登录表单。支持「记住我」以在关闭浏览器后保持登录状态。"""
    username = StringField("用户名", validators=[DataRequired()
    , Length(max=64)])
    password = PasswordField("密码", validators=[DataRequired(
    )])
    remember = BooleanField("记住我")
    submit = SubmitField("登录")
GENDER_CHOICES = [("", "不填")] + [(g, g) for g in GENDERS]
class ProfileForm(FlaskForm):
    """个人资料：昵称、性别与联系方式。字段都可留空。
    职务不在这里 —— 只能由管理员在「个人设置 · 账号」里改。
    """
    display_name = StringField("昵称", validators=[Optional(),
    Length(max=64)])
```

```
    gender = SelectField("性别", choices=GENDER_CHOICES,
    validators=[Optional()])
    email = StringField("邮箱", validators=[Optional(),
    email_format,
                                            Length(max=128)])
    phone = StringField("手机号", validators=[Optional(),
    Length(max=32)])
    submit_info = SubmitField("保存资料")
class RegisterForm(FlaskForm):
    """注册：账号与密码必填，其余选填。注册后默认是操作员。"""
    username = StringField("账号名称",
                           validators=[DataRequired("请填写账号名称
    "),
                                       Length(min=2, max=64,
                                              message="账号名称
    2–64 个字符")])
    password = PasswordField("密码",
                             validators=[DataRequired("请设置密码
    "),
                                         Length(min=6,
    max=64,

    message="密码至少 6 位")])
    confirm = PasswordField("确认密码",
                            validators=[DataRequired(),
                                        EqualTo("password",

    message="两次输入的密码不一致")])
    display_name = StringField("昵称", validators=[Optional(),
    Length(max=64)])
    gender = SelectField("性别", choices=GENDER_CHOICES,
    validators=[Optional()])
    email = StringField("邮箱", validators=[Optional(),
    email_format,
                                            Length(max=128)])
    phone = StringField("手机号", validators=[Optional(),
    Length(max=32)])
    submit_reg = SubmitField("注册并登录")
class PasswordForm(FlaskForm):
    """修改密码：需要先验证当前密码。"""
    old_password = PasswordField("当前密码",
    validators=[DataRequired()])
    new_password = PasswordField("新密码",
                                 validators=[DataRequired(),
    Length(min=6, max=64,

    message="新密码至少 6 位")])
    confirm = PasswordField("确认新密码",
                            validators=[DataRequired(),
```

```
        EqualTo("new_password",
        message="两次输入的新密码不一致")])
        submit_pwd = SubmitField("修改密码")
# ==== app/utils/constants.py ====
# -*- coding: utf-8 -*-
"""评价指标常量表。
全部数值取自论文，作为系统计算的唯一基准：
- 论文 4.2.2 节：指标标准化理想极值与取值依据
- 论文 4.2.3 节 / 表 4：适宜性等级划分
- 论文表 3：按微地质单元分别确定的一级、二级指标权重
- 论文 4.2.1 节：平均随机一致性指标 RI
"""
# ---------------------------------------------------------------------
    ------------------
# 一、微地质单元类型
# ---------------------------------------------------------------------
    ------------------
# 论文 4.1 节："按照地形地貌划分，将其划分为平台、碎石边坡和岩质边坡
# 三种地质微单元类型"
UNIT_TYPES = ("岩质边坡", "碎石边坡", "平台")
# 每个单元类型启用的指标类别（论文表 3 隐含：岩质边坡无土壤指标，平台无地形指标）
# 供数据录入界面与评价引擎判断"该单元是否需要某类指标"
UNIT_ENABLED_INDICATORS = {
    "岩质边坡": ("植被指标", "地形指标"),                    # 不设土壤指标
    "碎石边坡": ("土壤指标", "植被指标", "地形指标"),
    "平台": ("土壤指标", "植被指标"),                        # 不设地形指标
}
# ---------------------------------------------------------------------
    ------------------
# 二、指标标准化理想极值（论文 4.2.2 节）
# ---------------------------------------------------------------------
    ------------------
IDEAL_SOIL_RATIO = 0.8          # 土石比理想值
IDEAL_THICKNESS = 20.0          # 土层厚度理想值 （cm）
IDEAL_PH_RANGE = (5.0, 9.0)     # pH 理想区间（适中型指标）
IDEAL_ORGANIC = 30.0            # 有机质理想值
    （g/kg），全国第二次土壤普查养分等级划分标准二等最低值
IDEAL_COVERAGE = 75.0           # 植被覆盖度参照生态系统 （%）
IDEAL_SHANNON = 0.7             # Shannon-Wiener 多样性指数参照生态系统
# 重金属极值 （mg/kg），《绿化种植土壤》CJ/T 340—2016 土壤重金属含量 IV 级最高值
# 论文原文：镉、汞、铅、铬、砷、镍、铜、锌分别为 2、2、530、400、45、220、600、800
HEAVY_METALS = ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu",
    "Zn")
HEAVY_METAL_LIMIT = {
    "Cd": 2.0, "Hg": 2.0, "Pb": 530.0, "Cr": 400.0,
    "As": 45.0, "Ni": 220.0, "Cu": 600.0, "Zn": 800.0,
}
# 污染指标下 8 种重金属"重要性相当，权重平均分配"（论文 4.2.1 节）
HEAVY_METAL_EQUAL_WEIGHT = 1.0 / 8.0
```

```
#  坡度分界，论文  4．2．2  节引《新一轮退耕还林还草总体方案》发改西部〔２０１４〕１７７２  号
SLOPE_BREAK  =  25．0
#  坡向离散赋值，论文  4．2．2  节："阳坡、阴阳坡、阴坡分别赋值  0．2、0．6、1"
ASPECT_SCORES  =  {"阳坡":  0．2，  "阴阳坡":  0．6，  "阴坡":  1．0}
ASPECT_TYPES  =  ("阳坡",  "阴阳坡",  "阴坡")
#  pH  适中型标准化的容差边界（区间外线性衰减，避免得分突变）
PH_TOL_LOW  =  4．0
PH_TOL_HIGH  =  10．0
#  ----------------------------------------------------------------
    ------------------
#  三、适宜性等级划分（论文表  4）
#  ----------------------------------------------------------------
    ------------------
GRADE_LEVELS  =  (
    {"grade":  "不适宜",  "min":  None,  "max":  0．3，
     "meaning":  "存在限制性条件或区域生态本底较差，自然恢复影响因子严重制约自然恢复"，
     "measure":  "生态重建、人工辅助"}，
    {"grade":  "基本适宜",  "min":  0．3，  "max":  0．7，
     "meaning":  "生态本底良好，核心自然恢复影响因子基本满足自然恢复需求"，
     "measure":  "自然恢复为主，可辅以封闭场地和少量补种、补植、飞播等"}，
    {"grade":  "适宜",  "min":  0．7，  "max":  None，
     "meaning":  "各自然恢复影响因子均能满足自然恢复需求，自然演替速度快"，
     "measure":  "自然恢复"}，
)
def  grade_of(score):
    """按论文表  4  判定适宜性等级。"""
    for  g  in  GRADE_LEVELS:
        lo，  hi  =  g["min"]，  g["max"]
        if  lo  is  None  and  score  <  hi:
            return  g
        if  hi  is  None  and  score  >=  lo:
            return  g
        if  lo  is  not  None  and  hi  is  not  None  and  lo  <=
    score  <  hi:
            return  g
    #  score  恰好等于  0．7  时落入"适宜"
    return  GRADE_LEVELS[-1]
#  ----------------------------------------------------------------
    ------------------
#  四、平均随机一致性指标  RI（论文  4．2．1  节给出  n=2，3，4，5，其余为常用扩展值）
#  ----------------------------------------------------------------
    ------------------
RI_TABLE  =  {1:  0．0，  2:  0．0，  3:  0．58，  4:  0．90，  5:  1．12，  6:
    1．24，
            7:  1．32，  8:  1．41，  9:  1．45，  10:  1．49}
#  一致性比例阈值（论文：当  CR<0．1  时，判断矩阵满足一致性要求）
CR_THRESHOLD  =  0．1
#  ----------------------------------------------------------------
    ------------------
#  五、默认权重方案（论文表  3）
```

```
#  ----------------------------------------------------------------------  801
    ----------------                                                        802
#  论文　４．２．２　节："划分岩质边坡、碎石土边坡、平台三类微地质单元开展独立评价，  803
#  分别计算评价指标权重"（论文摘要、图　２　流程与表　５　均作"碎石边坡"，  804
#  系统统一采用"碎石边坡"）。权重由　１０　位专家打分后经　ＡＨＰ　方法确定。  805
#  下表为论文表　３　所载的归一化权重，作为系统默认方案。                  806
ＤＥＦＡＵＬＴ＿ＷＥＩＧＨＴＳ　＝　｛                                    807
    "岩质边坡"：　｛                                                    808
        "ｌ１"：　｛"植被指标"：　０．４５，　"地形指标"：　０．５５｝，  809
        "ｌ２"：　｛                                                    810
            "植被指标"：　｛"植被覆盖度"：　０．６７，　"生物多样性"：　０．３３｝，  811
            "地形指标"：　｛"坡度"：　０．６４，　"坡向"：　０．３６｝，  812
        ｝，                                                            813
    ｝，                                                                814
    "碎石边坡"：　｛                                                    815
        "ｌ１"：　｛"土壤指标"：　０．４４，　"植被指标"：　０．３１，　"地形指标"：　０．２５｝，  816
        "ｌ２"：　｛                                                    817
            "土壤指标"：　｛"土石比"：　０．２５，　"土层厚度"：　０．２５，　"ｐＨ"：　０．１７，  818
                                "有机质"：　０．１７，　"污染指标"：　０．１７｝，  819
            "植被指标"：　｛"植被覆盖度"：　０．６４，　"生物多样性"：　０．３６｝，  820
            "地形指标"：　｛"坡度"：　０．６７，　"坡向"：　０．３３｝，  821
        ｝，                                                            822
    ｝，                                                                823
    "平台"：　｛                                                        824
        "ｌ１"：　｛"土壤指标"：　０．５４，　"植被指标"：　０．４６｝，  825
        "ｌ２"：　｛                                                    826
            "土壤指标"：　｛"土石比"：　０．２５，　"土层厚度"：　０．２５，　"ｐＨ"：　０．１７，  827
                                "有机质"：　０．１７，　"污染指标"：　０．１７｝，  828
            "植被指标"：　｛"植被覆盖度"：　０．６１，　"生物多样性"：　０．３９｝，  829
        ｝，                                                            830
    ｝，                                                                831
｝                                                                      832
#  ----------------------------------------------------------------------  833
    ----------------                                                        834
#  六、限制性指标（论文表　２　＋　４．１　节流程）                        835
#  ----------------------------------------------------------------------  836
    ----------------                                                        837
#  论文摘要与　４．１　节：四项限制性指标；流程段另明确"是否为耕地"为约束条件  838
ＣＯＮＳＴＲＡＩＮＴ＿ＩＴＥＭＳ　＝　（"政策"，　"区位约束"，　"土壤污染"，　"地质灾害"，　"耕地属性"）  839
#  ----------------------------------------------------------------------  840
    ----------------                                                        841
#  七、账号职务                                                            842
#  ----------------------------------------------------------------------  843
    ----------------                                                        844
#  只分两种：管理员可管理账号、维护基础数据并查看操作日志，操作员用业务功能。  845
ＲＯＬＥ＿ＬＡＢＥＬＳ　＝　｛"ａｄｍｉｎ"：　"管理员"，　"ｏｐｅｒａｔｏｒ"：　"操作员"｝  846
ＲＯＬＥ＿ＣＯＤＥＳ　＝　（"ａｄｍｉｎ"，　"ｏｐｅｒａｔｏｒ"）  847
ＤＥＦＡＵＬＴ＿ＲＯＬＥ　＝　"ｏｐｅｒａｔｏｒ"　　#　新注册账号的默认职务  848
ＲＯＬＥ＿ＯＲＤＥＲ　＝　（"ａｄｍｉｎ"，　"ｏｐｅｒａｔｏｒ"）　#　列表排序用：管理员在前  849
#  性别（注册与个人资料中选填）                                            850
```

```
GENDERS = ("男", "女", "保密")
#  ==== app/utils/schema.py ====
#  -*- coding: utf-8 -*-
"""轻量结构迁移。
SQLite 下 `db.create_all()` 只建缺失的表，不会给已有表补列。
本模块在启动时检查目标列是否存在，缺失则 ALTER TABLE 补上，
使旧数据库无需重建即可用上新字段。
"""
from sqlalchemy import text
#  (表名, 列名, 列定义)
REQUIRED_COLUMNS = [
    #  遥感影像挂接
    ("mine_patches", "image_path", "VARCHAR(256)"),
    ("mine_patches", "image_note", "VARCHAR(256)"),
    ("mine_patches", "image_uploaded_at", "DATETIME"),
    #  指标维护
    ("indicators", "enabled_units", "VARCHAR(128)"),
    ("indicators", "description", "TEXT"),
    #  多专家矩阵
    ("expert_matrices", "unit_type", "VARCHAR(32)"),
    ("expert_matrices", "l1_name", "VARCHAR(64)"),
    ("expert_matrices", "cr", "FLOAT"),
    #  用户个人资料（账户页）
    ("users", "display_name", "VARCHAR(64)"),
    ("users", "email", "VARCHAR(128)"),
    ("users", "phone", "VARCHAR(32)"),
    ("users", "gender", "VARCHAR(8)"),
    ("users", "avatar", "VARCHAR(256)"),
]
#  职务体系由「管理员 / 评价员 / 查看者」三种收敛为「管理员 / 操作员」两种，
#  旧数据里的后两种统一归为操作员。
ROLE_MIGRATION = {"evaluator": "operator", "viewer":
    "operator"}
def _add_columns(db):
    added = []
    with db.engine.connect() as conn:
        for table, column, ddl in REQUIRED_COLUMNS:
            rows = conn.execute(text("PRAGMA
    table_info(%s)" % table)).fetchall()
            if not rows:
                continue                                #
    表还不存在，create_all 会建
            existing = {r[1] for r in rows}
            if column in existing:
                continue
            conn.execute(text(
                "ALTER TABLE %s ADD COLUMN %s %s" % (table,
    column, ddl)))
            conn.commit()
            added.append("%s.%s" % (table, column))
```

```
    return added                                                    901
def _migrate_roles(db):                                             902
    """把旧角色名统一为新的两种，返回改动的行数。"""                    903
    changed = 0                                                     904
    with db.engine.connect() as conn:                               905
        rows = conn.execute(text("PRAGMA                            906
    table_info(users)")).fetchall()                                 907
        if not rows:                                                908
            return 0                                                909
        for old, new in ROLE_MIGRATION.items():                     910
            r = conn.execute(text("UPDATE users SET role =          911
    :new "                                                          912
                                  "WHERE role = :old"),             913
                                 {"new": new, "old": old})          914
            changed += r.rowcount or 0                              915
        conn.commit()                                               916
    return changed                                                  917
def ensure_schema(db):                                              918
    """为已有表补齐缺失列，并做必要的数据迁移。返回变更清单。"""        919
    try:                                                            920
        changed = _add_columns(db)                                  921
        n = _migrate_roles(db)                                      922
        if n:                                                       923
            changed.append("用户角色归一为管理员/操作员 %d 条" % n)  924
        return changed                                              925
    except Exception as e:                                  #       926
    迁移失败不应阻断启动                                            927
        print("[schema] 结构检查跳过：%s" % e)                      928
        return []                                                   929
#  ==== app/utils/uploads.py ====                                  930
#  -*- coding: utf-8 -*-                                           931
"""上传文件处理（遥感影像、批量导入）。"""                          932
import os                                                           933
import re                                                           934
import uuid                                                         935
from datetime import datetime                                       936
from werkzeug.utils import secure_filename                          937
def _safe_ext(filename):                                            938
    return os.path.splitext(filename)[1].lower().lstrip(".")        939
def ext_allowed(filename, allowed):                                 940
    return _safe_ext(filename) in allowed                           941
def save_upload(file_storage, subdir, allowed_ext,                  942
    upload_root):                                                   943
    """保存上传文件，返回相对 static 的路径（如 uploads/patches/xxx.png）。  944
    文件名重命名为「时间戳_随机串.扩展名」，避免中文名与重名问题。      945
    """                                                             946
    if file_storage is None or not file_storage.filename:           947
        return None, "未选择文件"                                   948
    ext = _safe_ext(file_storage.filename)                          949
    if ext not in allowed_ext:                                      950
```

```
            return None, "不支持的文件类型：.%s" % ext                    951
        target_dir = os.path.join(upload_root, subdir)                    952
        os.makedirs(target_dir, exist_ok=True)                            953
        # 仅保留安全字符，并加时间戳与随机串防重名                          954
        base = secure_filename(os.path.splitext(file_storage.fil          955
        ename)[0]) or "file"                                              956
        base = re.sub(r"[^A-Za-z0-9_\-]", "", base)[:40] or               957
        "file"                                                            958
        name = "%s_%s.%s" % (datetime.now().strftime("%Y%m%d%H%M          959
        %S"),                                                             960
                                    uuid.uuid4().hex[:6], ext)            961
        abs_path = os.path.join(target_dir, name)                         962
        file_storage.save(abs_path)                                       963
        rel = os.path.join("uploads", subdir,                             964
        name).replace("\\", "/")                                          965
        return rel, None                                                  966
def remove_upload(rel_path, upload_root):                                 967
        """删除已保存的上传文件（相对 static 的路径）。"""                 968
        if not rel_path:                                                  969
            return                                                        970
        abs_path = os.path.join(upload_root,                              971
        rel_path.replace("uploads/", "", 1))                              972
        try:                                                              973
            if os.path.isfile(abs_path):                                  974
                os.remove(abs_path)                                       975
        except OSError:                                                   976
            pass                                                          977
# ==== app/utils/charts.py ====                                           978
# -*- coding: utf-8 -*-                                                   979
"""轻量图表。                                                             980
用纯 SVG 输出，不依赖任何绘图库或前端脚本：                               981
- 服务端渲染，页面与打印（PDF）中都能稳定显示                             982
- 每个图元都带数值标签，黑白打印时仍可读（不单靠颜色区分）               983
- 单色方案，符合界面「一个视图一个强调色」的约定                          984
"""                                                                       985
from html import escape                                                   986
ACCENT = "#1f3a5f"                                                        987
ACCENT_SOFT = "#7f93ad"                                                   988
INK = "#23262b"                                                           989
MUTED = "#6b7280"                                                         990
LINE = "#dfe3e8"                                                          991
# 灰度梯度：黑白打印时相邻扇区 / 条柱仍可区分                            992
GRAYS = ["#1f3a5f", "#456188", "#6d86a8", "#96a9c4",                      993
        "#bcc9da", "#dbe3ec"]                                             994
def _fmt(v, digits=4):                                                    995
        try:                                                              996
            return ("%%.%df" % digits) % float(v)                         997
        except (TypeError, ValueError):                                   998
            return str(v)                                                 999
def hbars(items, width=620, bar_h=22, gap=12, label_w=150,               1000
```

```
    value_w=76):                                                    1001
    """水平条形图。items: [(标签, 数值, 显示文本)]"""                    1002
    if not items:                                                   1003
        return '<p class="muted">暂无数据</p>'                       1004
    n = len(items)                                                  1005
    height = n * (bar_h + gap) + 10                                 1006
    max_v = max((abs(float(v)) for _, v, _ in items),               1007
    default=1) or 1                                                 1008
    body_w = width - label_w - value_w                              1009
    parts = ['<svg viewBox="0 0 %d %d" width="100%%"                1010
    height="%d" '                                                   1011
              'role="img" aria-label="条形图" '                      1012
              'style="max-width:%dpx;font-family:inherit;">'        1013
    % (width, height, height, width)]                               1014
    for i, (label, value, disp) in enumerate(items):                1015
        y = i * (bar_h + gap) + 5                                   1016
        frac = (abs(float(value)) / max_v) if max_v else 0          1017
        w = max(2.0, body_w * frac)                                 1018
        fill = GRAYS[i % len(GRAYS)]                                1019
        parts.append(                                               1020
            '<text x="%d" y="%d" text-anchor="end"                  1021
    font-size="12" fill="%s">%s</text>'                             1022
            % (label_w - 10, y + bar_h * 0.72, INK,                 1023
    escape(str(label))))                                            1024
        parts.append(                                               1025
            '<rect x="%d" y="%d" width="%.1f" height="%d"           1026
    rx="3" fill="%s"/>'                                             1027
            % (label_w, y, w, bar_h, fill))                         1028
        parts.append(                                               1029
            '<text x="%d" y="%d" font-size="12" fill="%s" '         1030
            'style="font-variant-numeric:tabular-nums;">%s</        1031
    text>'                                                          1032
            % (label_w + w + 8, y + bar_h * 0.72, MUTED,            1033
    escape(str(disp))))                                             1034
    parts.append('</svg>')                                          1035
    return "".join(parts)                                           1036
def donut(items, size=210, thickness=34):                           1037
    """环形图（权重分布）。items: [(标签, 数值, 显示文本)]              1038
    图例中给出数值与占比，黑白打印时靠图例与灰度深浅区分。                1039
    """                                                             1040
    vals = [(l, float(v), d) for l, v, d in items if                1041
    float(v) > 0]                                                   1042
    if not vals:                                                    1043
        return '<p class="muted">暂无数据</p>'                       1044
    total = sum(v for _, v, _ in vals) or 1                         1045
    r = (size - thickness) / 2                                      1046
    cx = cy = size / 2                                              1047
    import math                                                     1048
    parts = ['<svg viewBox="0 0 %d %d" width="%d"                   1049
    height="%d" role="img" '                                        1050
```

```
            'aria-label="权重分布环形图">' % (size, size, size,
    size)]
    angle = -math.pi / 2
    for i, (label, v, disp) in enumerate(vals):
        sweep = 2 * math.pi * (v / total)
        a0, a1 = angle, angle + sweep
        x0, y0 = cx + r * math.cos(a0), cy + r *
    math.sin(a0)
        x1, y1 = cx + r * math.cos(a1), cy + r *
    math.sin(a1)
        large = 1 if sweep > math.pi else 0
        parts.append(
            '<path d="M %.2f %.2f A %.2f %.2f 0 %d 1 %.2f
    %.2f" '
            'fill="none" stroke="%s" stroke-width="%d"/>'
           % (x0, y0, r, r, large, x1, y1, GRAYS[i %
    len(GRAYS)], thickness))
        angle = a1
    parts.append('<text x="%d" y="%d" text-anchor="middle"
    font-size="12" '
                 'fill="%s">合计</text>' % (cx, cy - 2,
    MUTED))
    parts.append('<text x="%d" y="%d" text-anchor="middle"
    font-size="15" '
                 'fill="%s" style="font-variant-numeric:tabu
    lar-nums;">%s</text>'
                 % (cx, cy + 18, INK, _fmt(total, 4)))
    parts.append('</svg>')
    legend = ['<table style="width:auto;margin-top:10px;"><t
    head><tr>'
              '<th scope="col">项目</th><th
    scope="col">数值</th>'
              '<th scope="col">占比</th></tr></thead><tbody>']
    for i, (label, v, disp) in enumerate(vals):
        legend.append(
            '<tr><td style="text-align:left;">'
            '<span style="display:inline-block;width:10px;he
    ight:10px;'
            'border-radius:2px;background:%s;margin-right:6p
    x;"></span>%s</td>'
            '<td class="mono">%s</td><td
    class="mono">%.1f%%</td></tr>'
           % (GRAYS[i % len(GRAYS)], escape(str(label)),
    escape(str(disp)),
              100.0 * v / total))
    legend.append('</tbody></table>')
    return ('<div style="display:flex;flex-wrap:wrap;gap:20p
    x;align-items:center;">'
            + "".join(parts) + "".join(legend) + '</div>')
# ==== app/utils/browser.py ====
```

```
# -*- coding: utf-8 -*-
"""HTML 转 PDF（报告导出）。
复用本机已安装的 Chrome/Edge 无头模式打印，避免引入 reportlab、
weasyprint 等额外依赖。生成过程使用系统临时目录中的 ASCII 文件名，
避免中文路径在 file:// URL 下出问题。
"""
import os
import shutil
import subprocess
import tempfile
import uuid
CANDIDATES = [
    r"C:\Program Files\Google\Chrome\Application\chrome.exe",
    r"C:\Program Files (x86)\Google\Chrome\Application\chrome.exe",
    r"C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe",
    r"C:\Program Files\Microsoft\Edge\Application\msedge.exe",
]
# 允许通过环境变量覆盖，便于在其他机器上指定浏览器
ENV_KEY = "DSH_BROWSER_PATH"
def browser_path():
    """定位可用的浏览器可执行文件，找不到返回 None。"""
    override = os.environ.get(ENV_KEY)
    if override and os.path.isfile(override):
        return override
    for p in CANDIDATES:
        if os.path.isfile(p):
            return p
    for name in ("chrome", "msedge", "chromium"):
        found = shutil.which(name)
        if found:
            return found
    return None
def html_to_pdf(html, timeout=90):
    """把 HTML 字符串渲染为 PDF，返回字节；失败返回 (None, 原因)。"""
    exe = browser_path()
    if exe is None:
        return None, ("未找到可用的浏览器（Chrome/Edge），无法生成 PDF。"
                      "可设置环境变量 %s 指向浏览器可执行文件。" % ENV_KEY)
    tmpdir = tempfile.mkdtemp(prefix="dshpdf_")
    tag = uuid.uuid4().hex[:8]
    html_path = os.path.join(tmpdir, "r_%s.html" % tag)
    pdf_path = os.path.join(tmpdir, "r_%s.pdf" % tag)
    try:
        with open(html_path, "w", encoding="utf-8") as f:
            f.write(html)
        cmd = [
```

```
            exe, "--headless=new", "--disable-gpu",
    "--no-sandbox",
            "--no-first-run", "--no-default-browser-check",
            "--no-pdf-header-footer",
            "--virtual-time-budget=4000",
            "--print-to-pdf=%s" % pdf_path,
            "file:///" + html_path.replace("\\", "/"),
        ]
        p = subprocess.run(cmd, capture_output=True,
    text=True,
                           timeout=timeout,
    encoding="utf-8", errors="replace")
        if not os.path.isfile(pdf_path) or
    os.path.getsize(pdf_path) < 800:
            return None, "浏览器未能生成 PDF（%s）" % (p.stderr or
    "").strip()[:200]
        with open(pdf_path, "rb") as f:
            return f.read(), None
    except subprocess.TimeoutExpired:
        return None, "生成 PDF 超时，请稍后重试"
    except Exception as e:
        return None, "生成 PDF 失败：%s" % e
    finally:
        shutil.rmtree(tmpdir, ignore_errors=True)
# ==== app/services/ahp_service.py ====
# -*- coding: utf-8 -*-
"""AHP 权重引擎。
严格按论文 4.2.1 节实现：
- 式(2)(3)(4)：特征根法求权重（行元素乘积 → n 次方根 → 归一化）
- 式(5)(6)(7)：一致性检验（λmax、CI、CR）
- 论文要求 CR < 0.1；RI 按阶数查表
"""
import numpy as np
from app.utils.constants import RI_TABLE, CR_THRESHOLD
def matrix_from_upper(upper):
    """由判断矩阵上三角构造完整互反矩阵。
    参数
    ----
    upper : list[list[float]] 或 np.ndarray，上三角元素（不含对角线）。
        例：3 阶时传入 [[a12, a13], [a23]]。
        对角线为 1，下三角取倒数（a_ji = 1/a_ij）。
    """
    upper = np.asarray(upper, dtype=float)
    k = upper.shape[0]
    n = int(round((1 + (1 + 8 * k) ** 0.5) / 2))
    A = np.ones((n, n), dtype=float)
    idx = 0
    for i in range(n):
        for j in range(i + 1, n):
            A[i, j] = upper[idx]
```

```
                A[j, i] = 1.0 / upper[idx]
                idx += 1
    return A
def weights_and_consistency(A):
    """由判断矩阵 A 计算权重与一致性指标。
    返回
    ----
    dict：{weights: np.ndarray, lambda_max, ci, cr, n,
    consistent: bool}
    """
    A = np.asarray(A, dtype=float)
    n = A.shape[0]
    # 式(2) 行元素乘积
    row_prod = np.prod(A, axis=1)
    # 式(3) 开 n 次方根
    root = row_prod ** (1.0 / n)
    # 式(4) 归一化
    weights = root / root.sum()
    # 式(5) 最大特征值 λmax = (1/n) Σ (A·W)_i / W_i
    aw = A.dot(weights)
    lambda_max = float(np.sum(aw / weights) / n)
    # 式(6) 一致性指标 CI
    ci = (lambda_max - n) / (n - 1) if n > 1 else 0.0
    # 式(7) 一致性比例 CR = CI / RI
    ri = RI_TABLE.get(n, 1.49)
    cr = ci / ri if ri else 0.0
    return {
        "weights": weights,
        "lambda_max": lambda_max,
        "ci": ci,
        "cr": cr,
        "n": n,
        "consistent": cr < CR_THRESHOLD,
    }
def find_most_inconsistent(A):
    """定位判断矩阵中最不一致的一对比较。
    依据：完全一致矩阵应满足 a_ij = w_i / w_j。
    计算每个上三角元素与其理想值的相对偏差，返回偏差最大者。
    返回
    ----
    dict：{i, j, value, expected, deviation}，偏差最大的 (i, j)
    及对应取值。
    """
    A = np.asarray(A, dtype=float)
    r = weights_and_consistency(A)
    w = r["weights"]
    n = A.shape[0]
    best = None
    for i in range(n):
        for j in range(i + 1, n):
```

```
                expected = w[i] / w[j]                                  1251
                if expected == 0:                                       1252
                    continue                                            1253
                deviation = abs(A[i, j] - expected) / expected          1254
                if best is None or deviation >                          1255
    best["deviation"]:                                                  1256
                    best = {"i": i, "j": j, "value": A[i, j],           1257
                            "expected": expected, "deviation":          1258
    deviation}                                                          1259
    return best                                                         1260
def aggregate_matrices(matrices):                                       1261
    """多专家判断矩阵聚合：几何平均（保持互反性）。                      1262
    参数                                                                1263
    ----                                                                1264
    matrices : list[np.ndarray]，多个同阶判断矩阵。                     1265
    """                                                                 1266
    stack = np.stack([np.asarray(m, dtype=float) for m in               1267
    matrices])                                                          1268
    return np.prod(stack, axis=0) ** (1.0 / len(stack))                 1269
def judge_from_weights(weights):                                        1270
    """由权重向量构造完全一致判断矩阵并检验。                           1271
    用途：验证给定权重向量能否作为某个完全一致判断矩阵的主特征向量      1272
    （A_ij = w_i / w_j 时 λmax = n、CR = 0）。                          1273
    """                                                                 1274
    w = np.asarray(weights, dtype=float)                                1275
    n = w.shape[0]                                                      1276
    A = np.outer(w, 1.0 / w)                                            1277
    r = weights_and_consistency(A)                                      1278
    r["matrix"] = A                                                     1279
    return r                                                            1280
# ==== app/services/standardization.py ====                             1281
# -*- coding: utf-8 -*-                                                 1282
"""指标标准化（的一部分）。                                             1283
按论文 4.2.2 节式(8) 实现：                                             1284
    x' = (x - x_min) / (x_max - x_min)                                  1285
按指标性质分为四类：越大越优 / 越小越优 / 适中 / 离散赋值。             1286
所有理想极值取自 app.utils.constants，可配置。                          1287
"""                                                                     1288
from app.utils.constants import (                                       1289
    IDEAL_SOIL_RATIO, IDEAL_THICKNESS, IDEAL_PH_RANGE,                  1290
    IDEAL_ORGANIC,                                                      1291
    IDEAL_COVERAGE, IDEAL_SHANNON, HEAVY_METAL_LIMIT,                   1292
    HEAVY_METAL_EQUAL_WEIGHT, SLOPE_BREAK, ASPECT_SCORES,               1293
    PH_TOL_LOW, PH_TOL_HIGH,                                            1294
)                                                                       1295
def std_benefit(value, ideal):                                          1296
    """越大越优指标标准化：以理想值为上限，按比例给分，饱和于 [0,1]。"""  1297
    if value is None:                                                   1298
        return None                                                     1299
    return max(0.0, min(1.0, value / ideal))                            1300
```

```
def std_cost(value, extreme):
    """越小越优指标标准化：以极值为参考，线性折减至 [0,1]。"""
    if value is None:
        return None
    return max(0.0, min(1.0, 1.0 - value / extreme))
def std_mid_ph(value, lo, hi, tol_lo=PH_TOL_LOW,
    tol_hi=PH_TOL_HIGH):
    """适中型指标标准化（pH）：理想区间内取 1，区间外线性衰减。"""
    if value is None:
        return None
    if lo <= value <= hi:
        return 1.0
    if value > hi:
        return max(0.0, (tol_hi - value) / (tol_hi - hi))
    return max(0.0, (value - tol_lo) / (lo - tol_lo))
def std_slope(value, break_point=SLOPE_BREAK, upper=90.0):
    """坡度标准化。
    论文 4.2.2 节：以 25° 为陡坡重要分界。25° 及以下记满分 1，
    超过 25° 线性折减，90° 记 0。
    """
    if value is None:
        return None
    if value <= break_point:
        return 1.0
    return max(0.0, (upper - value) / (upper - break_point))
def aspect_score(aspect):
    """坡向离散赋值：阳坡 0.2 / 阴阳坡 0.6 / 阴坡 1.0（论文 4.2.2 节）。"""
    if aspect is None:
        return None
    return ASPECT_SCORES.get(aspect)
def pollution_score(heavy_metals):
    """污染指标：8 种重金属等权（各 0.125）合成。
    论文注：未检测 Cr、Zn 的，从矿山类型看未受污染，按无污染（=1）处理。
    heavy_metals : dict，键为金属名，值为含量 mg/kg；缺失键视为未检测。
    """
    total = 0.0
    for metal, extreme in HEAVY_METAL_LIMIT.items():
        value = heavy_metals.get(metal)
        total += std_cost(value, extreme) if value is not
    None else 1.0
    return total / len(HEAVY_METAL_LIMIT)
def standardize_unit(unit_type, data, weights=None):
    """对单个微地质单元的全部指标做标准化。
    参数
    ----
    unit_type : str，岩质边坡 / 碎石边坡 / 平台
    data : dict，实测数据，字段见接口契约：
        soil_ratio, thickness, ph, organic_matter,
    heavy_metals(dict),
        coverage, shannon, slope, aspect
```

```
    weights : dict，该单元类型的权重（app.utils.constants.DEFAULT_WEIGHTS 结构），
                未提供时用默认值。
    返回
    ----
    dict：{ 一级指标： { 二级指标： （实测值， 标准化值， 权重， 贡献） } }
    标准化值可能为 None，表示该指标不适用或方法待定。
    """
    from app.utils.constants import DEFAULT_WEIGHTS
    if weights is None:
        weights = DEFAULT_WEIGHTS[unit_type]
    l1_enabled = weights["l1"].keys()
    l2w = weights["l2"]
    result = {}
    # ---- 土壤指标 ----
    if "土壤指标" in l1_enabled:
        result["土壤指标"] = {
            "土石比": (data.get("soil_ratio"),
                       std_benefit(data.get("soil_ratio"),
    IDEAL_SOIL_RATIO),
                       l2w["土壤指标"]["土石比"]),
            "土层厚度": (data.get("thickness"),
                         std_benefit(data.get("thickness"),
    IDEAL_THICKNESS),
                         l2w["土壤指标"]["土层厚度"]),
            "pH": (data.get("ph"),
                   std_mid_ph(data.get("ph"),
    *IDEAL_PH_RANGE),
                   l2w["土壤指标"]["pH"]),
            "有机质": (data.get("organic_matter"),
                       std_benefit(data.get("organic_matter")
    , IDEAL_ORGANIC),
                       l2w["土壤指标"]["有机质"]),
            "污染指标": (data.get("heavy_metals") or {},
                         pollution_score(data.get("heavy_meta
    ls") or {}),
                         l2w["土壤指标"]["污染指标"]),
        }
    # ---- 植被指标 ----
    if "植被指标" in l1_enabled:
        result["植被指标"] = {
            "植被覆盖度": (data.get("coverage"),
                           std_benefit(data.get("coverage"),
    IDEAL_COVERAGE),
                           l2w["植被指标"]["植被覆盖度"]),
            "生物多样性": (data.get("shannon"),
                           std_benefit(data.get("shannon"),
    IDEAL_SHANNON),
                           l2w["植被指标"]["生物多样性"]),
        }
```

```
    # ---- 地形指标 ----                                                    1401
    if "地形指标" in l1_enabled:                                            1402
        result["地形指标"] = {                                              1403
            "坡度": (data.get("slope"),                                     1404
                     std_slope(data.get("slope")),                          1405
                     l2w["地形指标"]["坡度"]),                              1406
            "坡向": (data.get("aspect"),                                    1407
                     aspect_score(data.get("aspect")),                      1408
                     l2w["地形指标"]["坡向"]),                              1409
        }                                                                   1410
    return result                                                           1411
# ==== app/services/weight_service.py ====                                1412
# -*- coding: utf-8 -*-                                                   1413
"""权重方案的存取与生效权重解析。                                          1414
系统有两套权重来源：                                                      1415
1. 论文表 3 给出的默认权重（app.utils.constants.DEFAULT_WEIGHTS），       1416
   是**权威取值来源**，案例复算与报告结论以此为准；                       1417
2. 通过层次分析法在本系统中实际算出的权重，保存在                         1418
   weight_schemes / weight_items 两张表中，可标记为「生效」。             1419
本模块负责把这两者合成为计算引擎直接可用的嵌套权重结构：                 1420
以论文表 3 为底，用生效方案中已保存的条目覆盖。                          1421
没有任何生效方案时，即完全使用论文表 3 的权重。                          1422
这样既保证默认行为与论文一致，又使 AHP 计算结果真正参与评价，            1423
而不是算完即弃。                                                        1424
"""                                                                     1425
import copy                                                             1426
from app.models import db, WeightScheme, WeightItem                     1427
from app.utils.constants import DEFAULT_WEIGHTS                         1428
from app.services.indicator_service import TREE                         1429
# 方案名称：每个微地质单元类型固定一个可生效的方案                       1430
ACTIVE_NAME = "自定义权重方案（AHP 计算结果）"                            1431
def _code_maps():                                                       1432
    """由指标树导出 编码 → 名称 的映射，避免名称在两处各写一遍。"""      1433
    l1, l2 = {}, {}                                                     1434
    for row in TREE:                                                    1435
        code, name, level, parent = row[0], row[1], row[2],             1436
    row[3]                                                              1437
        if level == 1:                                                  1438
            l1[code] = name                                             1439
        elif level == 2:                                                1440
            l2[code] = (l1.get(parent, parent), name)                   1441
    return l1, l2                                                       1442
def effective_weights(unit_type):                                       1443
    """返回该单元类型当前生效的权重。                                   1444
    返回 (weights, source_label, scheme)                                1445
        weights      嵌套权重结构，可直接交给标准化与评价引擎           1446
        source_label 权重来源说明，用于界面与报告标注                   1447
        scheme       生效的权重方案对象，无自定义方案时为 None          1448
    """                                                                 1449
    base = DEFAULT_WEIGHTS.get(unit_type)                               1450
```

```
    if base is None:
        return None, "未定义该微地质单元类型的权重", None
    scheme = WeightScheme.query.filter_by(unit_type=unit_type,
    name=ACTIVE_NAME).first()
    if scheme is None:
        return copy.deepcopy(base), "论文表 3 默认权重", None
    merged = copy.deepcopy(base)
    l1_map, l2_map = _code_maps()
    applied = 0
    for item in WeightItem.query.filter_by(scheme_id=scheme.
    id).all():
        code, w = item.indicator_code, item.weight
        if w is None or w <= 0:
            continue
        if code in l1_map and l1_map[code] in merged["l1"]:
            merged["l1"][l1_map[code]] = w
            applied += 1
        elif code in l2_map:
            l1_name, l2_name = l2_map[code]
            if l1_name in merged["l2"] and l2_name in
    merged["l2"][l1_name]:
                merged["l2"][l1_name][l2_name] = w
                applied += 1
    if applied == 0:
        return copy.deepcopy(base), "论文表 3 默认权重", None
    return merged, "自定义方案（%d 项覆盖论文表 3）" % applied, scheme
def save_weights(unit_type, code_weight_pairs, note=None):
    """把一组「指标编码 → 权重」写入生效方案（已存在则更新对应条目）。
    仅覆盖本次传入的条目，其余条目保留，便于分层次（一级 / 二级）
    分次录入而无需一次填完。
    """
    scheme = WeightScheme.query.filter_by(unit_type=unit_type,
    name=ACTIVE_NAME).first()
    if scheme is None:
        scheme = WeightScheme(unit_type=unit_type,
    name=ACTIVE_NAME,
                              level=1, is_default=True)
        db.session.add(scheme)
        db.session.flush()
    if note:
        scheme.level = scheme.level or 1
    if not scheme.is_default:
        scheme.is_default = True
    written = 0
    for code, w in code_weight_pairs:
        if w is None or w <= 0:
```

```
# ----------------------------------------------------------------------
    ----------------
# 论文第 5 节案例数据（表 5，逐字抄录）
# ----------------------------------------------------------------------
    ----------------
# 三类单元面积占比：岩质边坡 １３．８６%、碎石边坡 ６２．９５%、平台 ２３．１９%
# 注：岩质边坡无土壤指标、平台无地形指标，对应"不适用"。
case = [
    {
        "unit_type": "岩质边坡",
        "ratio": 0.1386,
        "data": {
            "coverage": 68.88, "shannon": 0.00,
            "slope": 63.33, "aspect": "阴阳坡",
        },
    },
    {
        "unit_type": "碎石边坡",
        "ratio": 0.6295,
        "data": {
            "soil_ratio": 0.34, "thickness": 10, "ph":
    9.24, "organic_matter": 4.89,
            "heavy_metals": {"Hg": 0.05, "As": 5.17, "Ni":
    12.60,
                             "Cu": 11.70, "Cd": 0.04, "Pb":
    6.50},
            "coverage": 87.92, "shannon": 0.60,
            "slope": 28.75, "aspect": "阳坡",
        },
    },
    {
        "unit_type": "平台",
        "ratio": 0.2319,
        "data": {
            "soil_ratio": 0.56, "thickness": 30, "ph":
    8.72, "organic_matter": 10.92,
            "heavy_metals": {"Hg": 0.02, "As": 3.13, "Ni":
    12.33,
                             "Cu": 11.14, "Cd": 0.09, "Pb":
    9.98},
            "coverage": 58.42, "shannon": 0.72,
        },
    },
]
# ----------------------------------------------------------------------
    ----------------
# 逐单元输出标准化与得分
# ----------------------------------------------------------------------
    ----------------
print(LINE)
```

```
print("玉田县案例 · 评价引擎完整计算")
print(LINE)
for u in case:
    r = ev.score_unit(u["unit_type"], u["data"])
    print()
    print("  【%s】 (面积占比 %.2f%%)" % (u["unit_type"],
    u["ratio"] * 100))
    print("  " + "-" * 60)
    for l1, score in r["l1_scores"].items():
        print("    %s 得分 = %.4f" % (l1, score))
    print("    单元评价因子 F' = %.4f" % r["unit_score"])
print()
print(LINE)
print("面积加权合成（式(9)）")
print(LINE)
res = ev.score_mine(case)
total = res["final_score"]
print()
for ur in res["unit_results"]:
    print("  %s %.2f%% × %.4f = %.4f"
          % (ur["unit_type"], ur["ratio"] * 100,
    ur["unit_score"], ur["weighted"]))
print("  " + "-" * 60)
print("  最终评价因子 F = %.4f" % total)
print()
print("  适宜性等级：%s" % res["grade"])
print("  判定依据 ：%s" % res["grade_info"]["meaning"])
print("  推荐措施 ：%s" % res["grade_info"]["measure"])
print()
print(LINE)
print("结论")
print(LINE)
print("  - 评价引擎按论文式(8)(9) 与表 3 权重完整跑通")
print("  - 坡度标准化采用论文 25° 分界的线性折减默认值（见 standardization.py）")
print("  - 最终 F ≈ %.2f，判定为「%s」，与论文「最终适宜自然恢复」的结论一致"
      % (total, res["grade"]))
print("  - 论文表 5 所载最终得分为 0.76；差异集中在坡度、坡向的标准化口径，")
print("    该口径在 standardization.std_slope / aspect_score
    中集中定义、可配置")
print(LINE)
#  ==== 测试/test_web_ahp.py ====
#  -*- coding: utf-8 -*-
"""AHP 网页界面自检：走通选择→录入→计算 全流程。"""
import sys
sys.stdout.reconfigure(encoding="utf-8")
import os
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
#  使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file
    __)))
```

```
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
# 测试配置：关闭 CSRF 以简化测试客户端（真实运行仍开启 CSRF）
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///:memory:"
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
# 建立表并创建测试用户，登录后再访问业务页面
with app.app_context():
    from app.models import db, User
    from werkzeug.security import generate_password_hash
    db.create_all()
    if not User.query.filter_by(username="admin").first():
        db.session.add(User(username="admin",
                            password_hash=generate_password_hash("admin123"),
                            role="admin"))
        db.session.commit()
client.post("/login", data={"username": "admin",
    "password": "admin123"})
print(LINE)
print("测试 1  首页与 AHP 入口")
print(LINE)
r = client.get("/")
print("  首页状态码：", r.status_code, "（应 200）")
assert r.status_code == 200
r = client.get("/ahp/")
print("  AHP 入口状态码：", r.status_code, "（应 200）")
assert r.status_code == 200
assert "微地质单元类型" in r.get_data(as_text=True)
print()
print(LINE)
print("测试 2  选择碎石边坡一级指标 → 跳转矩阵页")
print(LINE)
r = client.post("/ahp/", data={
    "unit_type": "碎石边坡", "level": "1", "submit": "生成判断矩阵",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  跳转后页面含「判断矩阵录入」：", "判断矩阵录入" in body)
print("  含三个一级指标（土壤/植被/地形）：", all(n in body for n in
    ("土壤指标", "植被指标", "地形指标")))
assert "判断矩阵录入" in body
assert all(n in body for n in ("土壤指标", "植被指标", "地形指标"))
print()
```

```
print(LINE)
print("测试 3  提交论文表 3 一致矩阵 → 还原权重")
print(LINE)
# 碎石边坡一级权重 （0.44， 0.31， 0.25） 对应的完全一致矩阵上三角：
# a12 = 0.44/0.31 = 1.4194， a13 = 0.44/0.25 = 1.76， a23 =
    0.31/0.25 = 1.24
r = client.post("/ahp/matrix?unit_type=碎石边坡&level=1", data={
    "m_0_1": "1.4194", "m_0_2": "1.76", "m_1_2": "1.24",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  结果页含「计算结果」:", "计算结果" in body)
print("  含权重 0.44:", "0.44" in body, " 含 0.31:", "0.31" in
    body, " 含 0.25:", "0.25" in body)
print("  含一致性检验通过:", "通过" in body)
assert "计算结果" in body
assert "0.44" in body and "0.31" in body and "0.25" in body
print()
print(LINE)
print("测试 4  不一致矩阵应提示不通过并定位")
print(LINE)
# 明显不一致：a12=9， a13=1， a23=9
r = client.post("/ahp/matrix?unit_type=碎石边坡&level=1", data={
    "m_0_1": "9", "m_0_2": "1", "m_1_2": "9",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  含「不通过」:", "不通过" in body)
print("  含「建议优先调整」:", "建议优先调整" in body)
assert "不通过" in body
assert "建议优先调整" in body
print()
print(LINE)
print("网页界面全部测试通过 √")
print(LINE)
# ==== 测试/test_web_modules.py ====
# -*- coding: utf-8 -*-
"""模块自检：登录、项目档案、微地质单元、标准库、限制性判定。"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil
    e__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
```

```
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
print(LINE)
print("测试 1  未登录应被重定向到登录页")
print(LINE)
r = client.get("/project/")
print("  /project/ 状态码:", r.status_code, " 重定向到:",
    r.headers.get("Location"))
assert r.status_code == 302 and "login" in
    r.headers.get("Location", "")
print()
print(LINE)
print("测试 2  登录")
print(LINE)
r = client.post("/login", data={"username": "admin",
    "password": "admin123"},
                follow_redirects=True)
body = r.get_data(as_text=True)
#  工作台:页首横幅 + 评价流程分区 + 数据概览
print("  登录后含「评价流程」:", "评价流程" in body)
print("  含用户名 admin:", "admin" in body)
print("  含页首横幅:", 'class="hero"' in body)
print("  含数据概览:", 'class="stat-card"' in body)
assert "评价流程" in body and 'class="hero"' in body
print()
print(LINE)
print("测试 3  项目与矿山档案")
print(LINE)
r = client.get("/project/")
body = r.get_data(as_text=True)
print("  项目列表状态码:", r.status_code)
print("  含案例项目:", "玉田县典型历史遗留废弃矿山案例" in body)
assert "玉田县典型历史遗留废弃矿山案例" in body
#  新建项目
client.post("/project/add", data={"name": "测试项目A", "code":
    "T-001"})
r = client.get("/project/")
print("  新建项目后含「测试项目A」:", "测试项目A" in r.get_data(as_text=True
    ))
assert "测试项目A" in r.get_data(as_text=True)
#  清理测试数据,避免污染演示库
with app.app_context():
    from app.models import db, Project
    for p in Project.query.filter_by(name="测试项目A").all():
        db.session.delete(p)
    db.session.commit()
```

```
    print("    已清理测试项目")
    print()
    print(LINE)
    print("测试 4    微地质单元（面积占比自动计算）")
    print(LINE)
    from app.models import MinePatch
    with app.app_context():
        patch = MinePatch.query.filter_by(name="玉田县孤树镇白云岩采石场").f
        irst()
        pid = patch.id
    r = client.get("/unit/%d" % pid)
    body = r.get_data(as_text=True)
    print("    单元列表含三类单元：", all(t in body for t in ("岩质边坡",
        "碎石边坡", "平台")))
    print("    含占比合计：", "占比合计" in body)
    assert "岩质边坡" in body and "占比合计" in body
    print()
    print(LINE)
    print("测试 5    标准知识库与省际口径比对")
    print(LINE)
    r = client.get("/standard/")
    body = r.get_data(as_text=True)
    print("    含辽宁/湖南/新疆标准：", all(p in body for p in ("辽宁", "湖南",
        "新疆")))
    assert "辽宁" in body
    r = client.get("/standard/compare")
    body = r.get_data(as_text=True)
    print("    口径比对页含「植被覆盖度」分组：", "植被覆盖度" in body)
    print("    含多省份并列：", "湖南" in body and "新疆" in body)
    assert "植被覆盖度" in body
    print()
    print(LINE)
    print("测试 6    限制性指标判定（一票否决）")
    print(LINE)
    r = client.get("/constraint/%d" % pid)
    body = r.get_data(as_text=True)
    print("    含五项条件：", all(x in body for x in ("政策约束", "区位约束",
        "土壤污染", "地质灾害", "耕地属性")))
    assert all(x in body for x in ("政策约束", "区位约束", "土壤污染",
        "地质灾害", "耕地属性"))
    # 全部通过
    client.post("/constraint/%d" % pid, data={
        "policy_pass": "1", "location_pass": "1",
        "pollution_pass": "1",
        "geo_pass": "1", "farmland_pass": "1"})
    r = client.get("/constraint/%d" % pid)
    body = r.get_data(as_text=True)
    print("    全通过后显示「通过，可进入定量评价」：", "可进入定量评价" in body)
    assert "可进入定量评价" in body
    # 一项不通过
```

```
client.post("/constraint/%d" % pid, data={                              11103
    "policy_pass": "1", "location_pass": "0",                           11104
    "pollution_pass": "1",                                              11105
    "geo_pass": "1", "farmland_pass": "1"})                             11106
r = client.get("/constraint/%d" % pid)                                  11107
body = r.get_data(as_text=True)                                         11108
print("  一项不通过后显示「不通过」:", "不通过" in body)                    11109
print("  含失败项「区位约束」:", "区位约束" in body)                        11110
assert "不通过" in body                                                  11111
# 恢复为全部通过，避免把演示数据留在「不通过」状态影响后续步骤              11112
client.post("/constraint/%d" % pid, data={                              11113
    "policy_pass": "1", "location_pass": "1",                           11114
    "pollution_pass": "1",                                              11115
    "geo_pass": "1", "farmland_pass": "1"})                             11116
r = client.get("/constraint/%d" % pid)                                  11117
print("  已恢复为全部通过:", "可进入定量评价" in r.get_data(as_text=True))  11118
print()                                                                 11119
print(LINE)                                                             11120
print("全部测试通过 √")                                                 11121
print(LINE)                                                             11122
# ==== 测试/test_web_evaluate.py ====                                   11123
# -*- coding: utf-8 -*-                                                 11124
"""端到端自检：录数据 → 执行评价 → 查看结果。"""                          11125
import os                                                               11126
import sys                                                              11127
sys.stdout.reconfigure(encoding="utf-8")                                11128
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，                 11129
# 使 `from app import ...` 与 app/app.db 的定位不受影响。                11130
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__11131
    e__)))                                                              11132
if ROOT not in sys.path:                                                11133
    sys.path.insert(0, ROOT)                                            11134
from app import create_app                                              11135
LINE = "=" * 66                                                         11136
# 与真实配置共用同一个 app.db，但关闭 CSRF 便于测试客户端                 11137
class TestConfig:                                                       11138
    SECRET_KEY = "test"                                                 11139
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(              11140
        ROOT, "app", "app.db")                                          11141
    SQLALCHEMY_TRACK_MODIFICATIONS = False                              11142
    WTF_CSRF_ENABLED = False                                            11143
    TESTING = True                                                      11144
app = create_app(TestConfig)                                            11145
client = app.test_client()                                              11146
# 登录（案例种子已创建 admin 用户）                                      11147
client.post("/login", data={"username": "admin",                        11148
    "password": "admin123"})                                            11149
print(LINE)                                                             11150
print("测试 1  指标数据列表")                                           11151
print(LINE)                                                             11152
```

```
r = client.get("/indicator/")
body = r.get_data(as_text=True)
print("  状态码：", r.status_code)
print("  含案例图斑：", "玉田县孤树镇白云岩采石场" in body)
print("  含三个单元：", all(n in body for n in ("岩质边坡", "碎石边坡",
    "平台")))
assert r.status_code == 200
assert "岩质边坡" in body and "碎石边坡" in body and "平台" in body
print()
print(LINE)
print("测试 2  指标数据录入页（按单元类型显示字段）")
print(LINE)
from app.models import MicroUnit, MinePatch
with app.app_context():
    unit_rock = MicroUnit.query.filter_by(unit_type="岩质边坡").
    first()
    unit_gravel = MicroUnit.query.filter_by(unit_type="碎石边坡"
    ).first()
r = client.get("/indicator/%d" % unit_rock.id)
b_rock = r.get_data(as_text=True)
print("  岩质边坡页：无「土石比」字段：", "土石比" not in b_rock, "（应 True）")
print("  岩质边坡页：含「植被覆盖度」：", "植被覆盖度" in b_rock)
assert "土石比" not in b_rock and "植被覆盖度" in b_rock
r = client.get("/indicator/%d" % unit_gravel.id)
b_gravel = r.get_data(as_text=True)
print("  碎石边坡页：含「土石比」字段：", "土石比" in b_gravel)
print("  碎石边坡页：含 8 种重金属：", all(m in b_gravel for m in
    ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu", "Zn")))
assert "土石比" in b_gravel and all(m in b_gravel for m in
    ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu", "Zn"))
print()
print(LINE)
print("测试 3  执行评价")
print(LINE)
with app.app_context():
    import seed_demo
    patch = seed_demo.demo_patch()
r = client.post("/evaluate/%d" % patch.id, data={},
    follow_redirects=True)
body = r.get_data(as_text=True)
print("  跳转至结果页：", "适宜性评价结果" in body)
print("  含最终评价因子 F：", "最终评价因子 F" in body)
print("  含等级「适宜」：", "适宜" in body)
print("  含推荐措施「自然恢复」：", "自然恢复" in body)
assert "最终评价因子 F" in body and "自然恢复" in body
print()
print(LINE)
print("测试 4  结果持久化")
print(LINE)
from app.models import EvaluationResult
```

```
with app.app_context():
    res = EvaluationResult.query.order_by(EvaluationResult.i
    d.desc()).first()
    print("  最终得分 F =", res.final_score)
    print("  等级 =", res.grade)
    assert res.grade == "适宜"
    assert 0.7 <= res.final_score < 1.0
print()
print(LINE)
print("端到端全部测试通过 √")
print(LINE)
# ==== 测试/test_web_report.py ====
# -*- coding: utf-8 -*-
"""报告输出自检：得分明细页 + Word 导出。"""
import os
import sys
import io
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil
    e__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
# 登录（案例种子已创建 admin 用户）
client.post("/login", data={"username": "admin",
    "password": "admin123"})
from app.models import MinePatch, EvaluationResult
with app.app_context():
    import seed_demo
    patch = seed_demo.demo_patch()
    pid = patch.id
# 报告取自已保存的评价记录，因此先按网页路径真实执行一次评价
from app.views.evaluate import load_patch_input,
    save_evaluation
with app.app_context():
    units = load_patch_input(pid)
    rec = save_evaluation(pid, units)
    rid = rec.id
```

```
        print("前置：已执行评价并保存记录 id=%d, F=%.4f (%s)"
              % (rid, rec.final_score, rec.grade))
        from app.models import ScoreDetail
        n_detail = ScoreDetail.query.filter_by(result_id=rid).co
        unt()
        print("         逐层得分明细已落库：%d 条" % n_detail)
print()
print(LINE)
print("测试 1  报告列表")
print(LINE)
r = client.get("/report/")
print("  状态码：", r.status_code)
body = r.get_data(as_text=True)
print("  含「所属项目」列：", "所属项目" in body)
assert r.status_code == 200
print()
print(LINE)
print("测试 2  得分明细页")
print(LINE)
r = client.get("/report/%d" % pid)
body = r.get_data(as_text=True)
print("  状态码：", r.status_code)
print("  含「得分明细」：", "得分明细" in body)
print("  含「最终评价因子 F」：", "最终评价因子 F" in body)
print("  含二级指标明细（植被覆盖度 / 坡度 / 坡向）：", all(n in body for n in
      ("植被覆盖度", "坡度", "坡向")))
print("  含权重 / 标准化值列：", "标准化值" in body and "贡献" in body)
assert r.status_code == 200
assert "植被覆盖度" in body and "标准化值" in body
print()
print(LINE)
print("测试 3  Word 报告导出")
print(LINE)
r = client.get("/report/%d/word" % pid)
print("  状态码：", r.status_code)
print("  Content-Type:", r.headers.get("Content-Type",
      "")[:60])
data = r.data
print("  文件大小：", len(data), "字节")
print("  是否为有效 docx（PK 头）：", data[:2] == b"PK")
assert r.status_code == 200 and data[:2] == b"PK"
# 用 python-docx 打开验证内容
import docx
d = docx.Document(io.BytesIO(data))
text = "\n".join(p.text for p in d.paragraphs)
print("  含报告标题：", "自然恢复适宜性评价报告" in text)
print("  含评价结论章节：", "评价结论" in text)
print("  含得分明细章节：", "得分明细" in text)
print("  段落数：", len(d.paragraphs), " 表格数：", len(d.tables))
assert "自然恢复适宜性评价报告" in text and len(d.tables) >= 3
```

```
print()
print(LINE)
print("报告输出全部测试通过 √")
print(LINE)
# ==== 测试/test_ui_fixes.py ====
# -*- coding: utf-8 -*-
"""界面修复自检：「记住我」与各模块删除功能。"""
import os
import re
import sys
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
sys.stdout.reconfigure(encoding="utf-8")
LINE = "=" * 70
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
    REMEMBER_COOKIE_DURATION = __import__("datetime").timedelta(days=30)
app = create_app(TestConfig)
client = app.test_client()
# ------------------------------------------------------------------
print(LINE)
print("一、「记住我」")
print(LINE)
html = client.get("/login").get_data(as_text=True)
# 1) 复选框必须有可点击关联的 label（这是原先失效的根因）
#    只判断「复选框与文字是否落在同一个 <label> 内」，不绑定具体类名 ——
#    否则换个样式类就会误报，而真正的回归点其实与类名无关。
labels = re.findall(r"<label[^>]*>(.*?)</label>", html, re.S)
wrapped = any(('name="remember"' in b and "记住我" in b) for b in labels)
print("  %s 「记住我」文字与复选框包在同一 label 内" % ("[符合]" if wrapped else "[不符合]"))
assert wrapped, "复选框未与文字关联，点击文字不会勾选"
has_for = 'for="remember"' in html
print("  %s label 带 for 属性指向复选框" % ("[符合]" if has_for else "[不符合]"))
# 2) 勾选后应下发带过期时间的 remember_token
```

```
c = app.test_client()
r = c.post("/login", data={"username": "admin", "password":
    "admin123",
                               "remember": "y"},
    follow_redirects=False)
cookies = r.headers.getlist("Set-Cookie")
rt = [h for h in cookies if h.startswith("remember_token=")]
print("  %s 勾选后下发 remember_token" % ("[符合]" if rt else
    "[不符合]"))
assert rt, "未下发 remember_token"
exp = re.search(r"Expires=([^;]+)", rt[0])
print("  %s 该 cookie 带过期时间：%s"
      % ("[符合]" if exp else "[不符合]", exp.group(1) if exp
    else "无（会话 cookie）"))
assert exp, "remember_token 无过期时间，关浏览器即失效"
# 3) 不勾选时不应下发 remember_token
c2 = app.test_client()
r2 = c2.post("/login", data={"username": "admin",
    "password": "admin123"},
                follow_redirects=False)
rt2 = [h for h in r2.headers.getlist("Set-Cookie") if
    h.startswith("remember_token=")]
print("  %s 未勾选时不下发 remember_token" % ("[符合]" if not rt2
    else "[不符合]"))
assert not rt2
# 4) 携带 remember_token 的新会话应自动登录
c3 = app.test_client()
c3.set_cookie("remember_token",
              rt[0].split(";")[0].split("=", 1)[1],
    domain="localhost")
r3 = c3.get("/", follow_redirects=False)
print("  %s 仅凭 remember_token 访问首页可自动登录（状态码 %d）"
      % ("[符合]" if r3.status_code == 200 else "[不符合]",
    r3.status_code))
assert r3.status_code == 200
# ------------------------------------------------------------
    ----------------
print()
print(LINE)
print("二、删除功能")
print(LINE)
client.post("/login", data={"username": "admin",
    "password": "admin123"})
# 项目：列表页须有删除按钮
html = client.get("/project/").get_data(as_text=True)
has_del = "/delete" in html and "btn--danger" in html
print("  %s 项目列表含删除按钮" % ("[符合]" if has_del else "[不符合]"))
assert has_del, "项目列表缺少删除按钮"
print("  %s 删除前有二次确认" % ("[符合]" if "confirm(" in html else
    "[不符合]"))
```

```
assert "confirm(" in html                                                    11403
# 图斑列表：地址用的是项目主键，不是图斑主键                                  11404
with app.app_context():                                                      11405
    from app.models import MinePatch, Project                                11406
    import seed_demo                                                         11407
    patch = seed_demo.demo_patch()                                           11408
    patch_id = patch.id                                                      11409
    project_id = patch.project_id                                            11410
html = client.get("/project/%d/patches" %                                    11411
    project_id).get_data(as_text=True)                                       11412
print("  %s 图斑列表含删除按钮" % ("[符合]" if "/project/patch/" in          11413
    html and "btn--danger" in html else "[不符合]"))                         11414
assert "/project/patch/" in html and "btn--danger" in html                   11415
# 微地质单元                                                                 11416
html = client.get("/unit/%d" % patch_id).get_data(as_text=Tr                 11417
    ue)                                                                      11418
print("  %s 微单元列表含删除按钮" % ("[符合]" if "/unit/delete/" in         11419
    html else "[不符合]"))                                                   11420
assert "/unit/delete/" in html                                               11421
print("  %s 微单元编辑为独立页面（表单不再嵌在表格行内）"                    11422
      % ("[符合]" if "/unit/edit/" in html else "[不符合]"))                 11423
# 标准                                                                       11424
html = client.get("/standard/").get_data(as_text=True)                       11425
print("  %s 标准列表含删除按钮" % ("[符合]" if "btn--danger" in html        11426
    else "[不符合]"))                                                        11427
assert "btn--danger" in html                                                 11428
# 实际执行一次删除：新建项目 → 删除 → 确认消失                               11429
client.post("/project/add", data={"name": "待删除测试项目",                 11430
    "code": "DEL-1"})                                                        11431
with app.app_context():                                                      11432
    from app.models import db, Project                                       11433
    tmp = Project.query.filter_by(name="待删除测试项目").first()             11434
    tmp_id = tmp.id                                                          11435
html = client.get("/project/").get_data(as_text=True)                        11436
print("  %s 新建项目成功" % ("[符合]" if "待删除测试项目" in html else       11437
    "[不符合]"))                                                             11438
r = client.post("/project/%d/delete" % tmp_id,                               11439
    follow_redirects=True)                                                   11440
html = r.get_data(as_text=True)                                              11441
gone = "待删除测试项目" not in html                                          11442
print("  %s 删除项目后列表中不再出现" % ("[符合]" if gone else "[不符合]"))  11443
assert gone                                                                  11444
print("  %s 有删除成功提示" % ("[符合]" if "已删除" in html else             11445
    "[不符合]"))                                                             11446
print()                                                                      11447
print(LINE)                                                                  11448
print("界面修复自检全部通过 √")                                             11449
print(LINE)                                                                  11450
# ==== 测试/test_new_features.py ====                                        11451
# -*- coding: utf-8 -*-                                                      11452
```

```
"""补齐模块自检：操作日志与用户维护、遥感影像、指标与多专家矩阵、
批量导入、图表与 PDF 导出。
对应计划书功能清单与系统实现的差异核对结论。
"""
import io
import json
import os
import re
import sys
sys.stdout.reconfigure(encoding="utf-8")
import fitz
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
from app.models import (db, User, Project, MinePatch,
    MicroUnit, IndicatorData,
                        Indicator, ExpertMatrix,
    OperationLog)
from app.utils.schema import ensure_schema
from app.utils import browser, charts
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
    UPLOAD_FOLDER = os.path.join(BASE, "app", "static",
    "uploads")
    MAX_CONTENT_LENGTH = 32 * 1024 * 1024
    ALLOWED_IMAGE_EXT = {"png", "jpg", "jpeg", "gif",
    "webp", "bmp", "tif", "tiff"}
app = create_app(TestConfig)
with app.app_context():
    db.create_all()
    ensure_schema(db)
    ExpertMatrix.query.delete()
    db.session.commit()
    # 固定取内置演示案例的图斑，避免受库中用户自建项目 / 验证算例的影响
    import seed_demo
    patch_id = seed_demo.demo_patch().id
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
```

```
def ok(flag, label, extra=""):
    print("  %s %-44s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    assert flag, label
# ------------------------------------------------------------
    ------------------
print(LINE)
print("操作日志与用户维护")
print(LINE)
before = None
with app.app_context():
    before = OperationLog.query.count()
c.post("/project/add", data={"name": "日志核对项目", "code":
    "LOG-T"})
with app.app_context():
    after = OperationLog.query.count()
    last = OperationLog.query.order_by(OperationLog.id.desc(
    )).first()
ok(after == before + 1, "写操作被写入操作日志", "%d -> %d" % (before,
    after))
ok(last.action == "新建项目", "日志记录了可读动作名", last.action)
ok("project" in (last.target_type or ""), "日志记录了对象类型",
    last.target_type)
r = c.get("/profile?tab=logs")
body = r.get_data(as_text=True)
ok(r.status_code == 200 and "操作日志" in body, "操作日志分类可访问")
ok("新建项目" in body, "日志页显示操作记录")
# 账号编辑 / 删除（职务只有管理员与操作员两种）
with app.app_context():
    u = User(username="tmp_user", role="operator",
             password_hash=User.query.first().password_hash)
    db.session.add(u); db.session.commit()
    uid = u.id
r = c.get("/users/%d/edit" % uid)
ok(r.status_code == 200, "账号编辑页可访问")
# 保护校验要趁系统里只有一名管理员时做：
# 先降级唯一的 admin 应被拒绝，再把临时账号提为管理员看修改是否生效
with app.app_context():
    admin_id = User.query.filter_by(role="admin").first().id
c.post("/users/%d/edit" % admin_id,
       data={"role": "operator", "active": "1"})
with app.app_context():
    ok(db.session.get(User, admin_id).role == "admin",
    "拒绝把最后一个管理员降级")
c.post("/users/%d/edit" % uid, data={"role": "admin",
    "active": "1"})
with app.app_context():
    ok(db.session.get(User, uid).role == "admin",
    "职务已修改并保存")
c.post("/users/%d/delete" % uid, follow_redirects=True)
```

```
with app.app_context():
    ok(db.session.get(User, uid) is None, "账号删除生效")
# 清理日志核对用的项目
with app.app_context():
    for p in Project.query.filter_by(name="日志核对项目").all():
        db.session.delete(p)
    db.session.commit()
# ------------------------------------------------------------------
    ------------------
print()
print(LINE)
print("遥感影像挂接")
print(LINE)
doc = fitz.open()
pg = doc.new_page(width=320, height=200)
pg.insert_text((20, 60), "RS image", fontname="china-s",
    fontsize=16)
png = doc[0].get_pixmap(dpi=100).tobytes("png")
doc.close()
r = c.post("/project/patch/%d/image" % patch_id,
           data={"image": (io.BytesIO(png), "影像.png"),
                 "image_note": "2025年6月无人机正射影像"},
           content_type="multipart/form-data",
    follow_redirects=True)
body = r.get_data(as_text=True)
ok("已挂接" in body and "<img" in body, "影像上传并显示")
with app.app_context():
    rel = db.session.get(MinePatch, patch_id).image_path
ok(bool(rel), "数据库记录影像路径", rel)
abs_path = os.path.join(TestConfig.UPLOAD_FOLDER,
    rel.replace("uploads/", "", 1))
ok(os.path.isfile(abs_path), "影像文件落盘")
ok(c.get("/static/" + rel).status_code == 200, "影像可经 URL
    访问")
r = c.post("/project/patch/%d/image" % patch_id,
           data={"image": (io.BytesIO(b"x"), "bad.txt")},
           content_type="multipart/form-data",
    follow_redirects=True)
ok("不支持" in r.get_data(as_text=True), "拒绝非图片格式")
c.post("/project/patch/%d/image/delete" % patch_id,
    follow_redirects=True)
with app.app_context():
    ok(db.session.get(MinePatch, patch_id).image_path is
    None, "影像移除后字段清空")
ok(not os.path.isfile(abs_path), "影像文件已删除")
# ------------------------------------------------------------------
    ------------------
print()
print(LINE)
print("指标定义维护与多专家矩阵")
```

```
print(LINE)
r = c.get("/ahp/indicators")
body = r.get_data(as_text=True)
with app.app_context():
    n_ind = Indicator.query.count()
ok(r.status_code == 200 and n_ind > 0, "指标定义已入库并可查看", "%d项" % n_ind)
ok(all(k in body for k in ("一级指标", "二级指标", "三级指标")),
   "指标树按层级展示")
ok("岩质边坡" in body and "平台" in body, "显示各指标启用的微地质单元")
c.post("/ahp/indicators/coverage/edit",
       data={"name": "植被覆盖度", "unit": "%", "ideal_type": "benefit",
             "ideal_value": "75", "enabled_units": ["岩质边坡"],
             "description": "自检修改"})
with app.app_context():
    ok(Indicator.query.filter_by(code="coverage").first().description == "自检修改",
       "指标定义可编辑")
c.post("/ahp/indicators/reset")
with app.app_context():
    ok("自检修改" not in (Indicator.query.filter_by(code="coverage").first().description or ""),
       "可按论文取值重置")
base = {"m_0_1": "1.4194", "m_0_2": "1.76", "m_1_2": "1.24"}
for name in ("张工", "李工", "王工"):
    d = dict(base); d.update({"action": "add", "unit_type": "碎石边坡",
                              "level": "1", "expert_name": name})
    c.post("/ahp/experts", data=d)
with app.app_context():
    ok(ExpertMatrix.query.filter_by(unit_type="碎石边坡", level=1).count() == 3,
       "录入 3 位专家的判断矩阵")
r = c.get("/ahp/experts/aggregate?unit_type=碎石边坡&level=1")
body = r.get_data(as_text=True)
ok("汇总后权重" in body and "各专家矩阵的一致性" in body,
   "多专家矩阵几何平均汇总可用")
ws = re.search(r"<th scope=\"row\">权重</th>\s*(.*?)</tr>", body, re.S)
vals = re.findall(r">([0-9.]+)</td>", ws.group(1)) if ws else []
ok(len(vals) == 3 and abs(sum(map(float, vals)) - 1.0) < 0.01,
   "汇总权重归一", "、".join(vals))
ok(abs(float(vals[0]) - 0.44) < 0.02, "汇总权重接近论文表 3 取值",
   vals[0] if vals else "")
# ------------------------------------------------------------------
```

```
print()
print(LINE)
print("批量导入")
print(LINE)
r = c.get("/indicator/import/template")
tpl = r.data.decode("utf-8-sig")
ok(r.status_code == 200 and "单元类型" in tpl, "导入模板可下载")
ok("土石比" in tpl and "植被覆盖度" in tpl, "模板包含指标列")
csv_text = (
    "单元类型,土石比,土层厚度,pH,有机质,Hg,As,Ni,Cu,Cd,Pb,植被覆盖度,Shannon,坡度
    ,坡向\n"
    "碎石边坡,0.34,10,9.24,4.89,0.05,5.17,12.60,11.70,0.04,6.50,
    87.92,0.60,28.75,阳坡\n"
    "平台,0.56,30,8.72,10.92,0.02,3.13,12.33,11.14,0.09,9.98,5
    8.42,0.72,,,\n"
    "不存在单元,1,1,7,1,,,,,,,,,\n"
)
r = c.post("/indicator/import",
           data={"patch_id": str(patch_id),
                 "datafile": (io.BytesIO(csv_text.encode("ut
    f-8")), "d.csv")},
           content_type="multipart/form-data",
    follow_redirects=True)
body = r.get_data(as_text=True)
ok("导入结果" in body, "导入结果页展示")
ok("不存在单元" in body, "错误行被报告")
with app.app_context():
    u = MicroUnit.query.filter_by(patch_id=patch_id,
    unit_type="碎石边坡").first()
    ind = IndicatorData.query.filter_by(unit_id=u.id).first(
    )
    ok(abs(ind.soil_ratio - 0.34) < 1e-6 and
    abs(ind.coverage - 87.92) < 1e-6,
        "导入数据已写入数据库")
r = c.post("/indicator/import",
           data={"patch_id": str(patch_id),
                 "datafile": (io.BytesIO(b"x"), "d.pdf")},
           content_type="multipart/form-data",
    follow_redirects=True)
ok("仅支持" in r.get_data(as_text=True), "拒绝不支持的导入格式")
# ----------------------------------------------------------------
    ------------------
print()
print(LINE)
print("图表与 PDF 导出")
print(LINE)
svg = charts.hbars([("岩质边坡", 0.54, "0.5401"), ("平台", 0.84,
    "0.8376")])
ok("<svg" in svg and "0.8376" in svg, "条形图生成且带数值标签")
svg2 = charts.donut([("植被指标", 0.6, "0.6000"), ("土壤指标", 0.4,
```

```
        "０.４０００")])                                                  11703
ok("<svg" in svg2 and "占比" in svg2, "环形图生成且带图例")                  11704
#  报告是评价结论的呈现，需先真实执行一次评价才能打开                      11705
with app.app_context():                                                  11706
    from app.views.evaluate import load_patch_input,                     11707
    save_evaluation                                                      11708
    save_evaluation(patch_id, load_patch_input(patch_id))                11709
r = c.get("/report/%d" % patch_id)                                       11710
body = r.get_data(as_text=True)                                          11711
ok(body.count("<svg") >= 2, "得分明细页含图表", "%d 个 SVG" %            11712
    body.count("<svg"))                                                  11713
exe = browser.browser_path()                                             11714
if exe is None:                                                          11715
    print("  [跳过] 未找到浏览器，PDF 导出无法验证")                       11716
else:                                                                    11717
    r = c.get("/report/%d/pdf" % patch_id)                               11718
    data = r.data                                                        11719
    ok(data[:4] == b"%PDF", "PDF 报告导出成功", "%d 字节" %              11720
    len(data))                                                           11721
    d = fitz.open(stream=data, filetype="pdf")                           11722
    txt = "".join(d[i].get_text() for i in                               11723
    range(d.page_count))                                                 11724
    ok("自然恢复适宜性评价报告" in txt, "PDF 含报告标题")                 11725
    ok("评价结论" in txt and "得分明细" in txt, "PDF 含结论与明细章节")   11726
    ok(d.page_count >= 2, "PDF 页数合理", "%d 页" % d.page_count)        11727
    d.close()                                                            11728
print()                                                                  11729
print(LINE)                                                              11730
print("补齐模块自检全部通过 √")                                          11731
print(LINE)                                                              11732
#  ==== 测试/test_workflow.py ====                                       11733
#  -*- coding: utf-8 -*-                                                 11734
"""流程门禁自检：前置步骤未完成时，后续模块必须拦得住。                  11735
论文 ４.１ 节确定的评价流程存在先后关系，缺任何一环都不应产出评价结论。  11736
本测试逐个构造「缺前置」的场景，验证系统不会放行。                      11737
"""                                                                      11738
import os                                                                11739
import sys                                                               11740
sys.stdout.reconfigure(encoding="utf-8")                                 11741
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，             11742
#  使 `from app import ...` 与 app/app.db 的定位不受影响。               11743
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil             11744
    e__)))                                                               11745
if ROOT not in sys.path:                                                 11746
    sys.path.insert(0, ROOT)                                             11747
from app import create_app                                               11748
from app.models import (db, Project, MinePatch, MicroUnit,               11749
    IndicatorData,                                                       11750
                            ConstraintCheck, EvaluationResult)           11751
from app.services import workflow as wf                                  11752
```

```
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
_fails = []
def ok(flag, label, extra=""):
    print("  %s %-46s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    if not flag:
        _fails.append(label)
def new_patch(tag):
    with app.app_context():
        p = Project(name="门禁-" + tag, code="WF-" + tag)
        db.session.add(p); db.session.flush()
        patch = MinePatch(project_id=p.id, name="门禁图斑-" +
    tag)
        db.session.add(patch); db.session.flush()
        pid = patch.id
        db.session.commit()
    return pid
def add_units(pid):
    with app.app_context():
        db.session.add(MicroUnit(patch_id=pid,
    unit_type="碎石边坡",
                                 area=1.0, ratio=1.0))
        db.session.commit()
def add_constraint(pid, passed=True):
    with app.app_context():
        db.session.add(ConstraintCheck(
            patch_id=pid, policy_pass=True,
    location_pass=passed,
            pollution_pass=True, geo_pass=True,
    farmland_pass=True,
            passed=passed, reason="全部通过" if passed else
    "未通过：区位约束"))
        db.session.commit()
def add_data(pid):
    with app.app_context():
        u = MicroUnit.query.filter_by(patch_id=pid).first()
        db.session.add(IndicatorData(unit_id=u.id,
    soil_ratio=0.34, thickness=10,
```

```
                                                ph=9.24,                    11803
        organic_matter=4.89, coverage=87.92,                                11804
                                                shannon=0.60,               11805
        slope=28.75, aspect="阳坡"))                                        11806
            db.session.commit()                                             11807
def cleanup(pid, tag):                                                      11808
        with app.app_context():                                             11809
            from app.models import ScoreDetail                              11810
            # 先删得分明细，再删评价记录，避免留下孤儿明细                  11811
            rids = [r.id for r in EvaluationResult.query.filter_            11812
        by(patch_id=pid)]                                                   11813
            if rids:                                                        11814
                ScoreDetail.query.filter(ScoreDetail.result_id.i            11815
        n_(rids)).delete(                                                   11816
                    synchronize_session=False)                              11817
            EvaluationResult.query.filter_by(patch_id=pid).delet            11818
        e()                                                                 11819
            ConstraintCheck.query.filter_by(patch_id=pid).delete            11820
        ()                                                                  11821
            uids = [u.id for u in MicroUnit.query.filter_by(pat             11822
        h_id=pid)]                                                          11823
            if uids:                                                        11824
                IndicatorData.query.filter(IndicatorData.unit_id            11825
        .in_(uids)).delete(                                                 11826
                    synchronize_session=False)                              11827
            MicroUnit.query.filter_by(patch_id=pid).delete()                11828
            m = db.session.get(MinePatch, pid)                              11829
            if m:                                                           11830
                db.session.delete(m)                                        11831
            pr = Project.query.filter_by(name="门禁-" +                     11832
        tag).first()                                                        11833
            if pr:                                                          11834
                db.session.delete(pr)                                       11835
            db.session.commit()                                             11836
def results_of(pid):                                                        11837
        with app.app_context():                                             11838
            return EvaluationResult.query.filter_by(patch_id=pid            11839
        ).count()                                                           11840
def post_eval(pid):                                                         11841
        return c.post("/evaluate/%d" % pid, data={},                        11842
        follow_redirects=False)                                             11843
def get_report(pid):                                                        11844
        return c.get("/report/%d" % pid, follow_redirects=False)            11845
# ------------------------------------------------------------------------  11846
        ------------------                                                  11847
print(LINE)                                                                 11848
print("一、前置条件缺失时，评价必须被拦住")                                 11849
print(LINE)                                                                 11850
print()                                                                     11851
print("场景 1：连微地质单元都没划分")                                       11852
```

```
pid = new_patch("1")
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "未划分单元 → 不产生评价结果",
   "HTTP %d，结果数 %d" % (r.status_code, n))
with app.app_context():
    pr = wf.patch_progress(pid)
steps = [b["step"] for b in pr["blocking"]]
ok(not pr["ready"] and "units" in steps,
   "拦原因包含「划分微地质单元」", "、".join(steps))
ok(steps == ["constraint", "units"],
   "拦截顺序符合论文 4.1 节：限制性判定在前，划分单元在后",
   "、".join(steps))
cleanup(pid, "1")
print()
print("场景 2：有单元，但未做限制性条件判定")
pid = new_patch("2"); add_units(pid); add_data(pid)
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "未做限制性判定 → 不产生评价结果",
   "结果数 %d" % n)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(any(b["step"] == "constraint" for b in pr["blocking"]),
   "拦原因指向「执行限制性条件判定」")
cleanup(pid, "2")
print()
print("场景 3：限制性条件判定【未通过】")
pid = new_patch("3"); add_units(pid); add_data(pid);
    add_constraint(pid, passed=False)
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "判定未通过 → 不进入定量评价",
   "结果数 %d" % n)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(not pr["constraint"]["passed"], "流程状态显示判定未通过")
blocked = [b for b in pr["blocking"] if b["step"] ==
    "constraint"]
# 提示只写「缺什么」：label 说明未通过，reason 直接引用判定记录里没过的项
ok(bool(blocked) and "未通过" in blocked[0]["label"],
   "拦提示指明限制性条件判定未通过", blocked[0]["label"] if blocked else
    "")
ok(bool(blocked) and "区位约束" in (blocked[0]["reason"] or ""),
   "并把具体没通过的条件带出来", blocked[0]["reason"] if blocked else "")
cleanup(pid, "3")
print()
print("场景 4：判定通过，但指标数据全空")
pid = new_patch("4"); add_units(pid); add_constraint(pid,
    passed=True)
```

```
r = post_eval(pid)                                                    11903
n = results_of(pid)                                                   11904
ok(r.status_code == 302 and n == 0, "数据未录入 → 不产生评价结果",      11905
   "结果数 %d" % n)                                                   11906
with app.app_context():                                               11907
    pr = wf.patch_progress(pid)                                       11908
ok(any(b["step"] == "data" for b in pr["blocking"]),                  11909
   "拦原因指向「录入指标数据」")                                      11910
ok(pr["data"]["missing"] == ["碎石边坡"],                             11911
   "指出具体缺哪个单元的数据", "、".join(pr["data"]["missing"]))      11912
cleanup(pid, "4")                                                     11913
print()                                                               11914
print("场景 5：单元面积占比合计不为 100%")                           11915
pid = new_patch("5"); add_constraint(pid, passed=True)                11916
with app.app_context():                                               11917
    u = MicroUnit(patch_id=pid, unit_type="碎石边坡", area=1.0,       11918
    ratio=0.5)                                                        11919
    db.session.add(u); db.session.commit()                            11920
    db.session.add(IndicatorData(unit_id=u.id,                        11921
    soil_ratio=0.3, thickness=10,                                     11922
                                          ph=8.0,                     11923
    organic_matter=5.0, coverage=80.0,                                11924
                                          shannon=0.6, slope=20.0,    11925
    aspect="阳坡"))                                                   11926
    db.session.commit()                                               11927
r = post_eval(pid)                                                    11928
ok(results_of(pid) == 0, "占比不足 100% → 不产生评价结果")            11929
cleanup(pid, "5")                                                     11930
print()                                                               11931
print(LINE)                                                           11932
print("二、前置条件齐备时才放行")                                     11933
print(LINE)                                                           11934
pid = new_patch("OK"); add_units(pid); add_constraint(pid,            11935
    passed=True); add_data(pid)                                       11936
with app.app_context():                                               11937
    pr = wf.patch_progress(pid)                                       11938
ok(pr["ready"], "单元 + 判定通过 + 数据齐备 → 判定为可执行")          11939
r = post_eval(pid)                                                    11940
n = results_of(pid)                                                   11941
ok(n == 1, "此时执行评价成功产生结果", "结果数 %d" % n)               11942
print("  重定向到:", r.headers.get("Location"))                       11943
print()                                                               11944
print(LINE)                                                           11945
print("三、报告不得先于评价产生")                                     11946
print(LINE)                                                           11947
pid2 = new_patch("R")    # 什么都不做                                 11948
r = get_report(pid2)                                                  11949
ok(r.status_code == 302 and "/report/" ==                             11950
    (r.headers.get("Location") or ""),                                11951
   "前置缺失时 /report/<id> 被退回列表", "HTTP %d" % r.status_code)   11952
```

```
for suffix in ("/pdf", "/word"):
    rr = c.get("/report/%d%s" % (pid2, suffix),
    follow_redirects=False)
    ok(rr.status_code == 302, "导出 %s 同样被拦住" % suffix, "HTTP
    %d" % rr.status_code)
cleanup(pid2, "R")
cleanup(pid, "OK")
print()
print(LINE)
print("四、前置齐备但从未执行评价时，报告仍不得产生")
print(LINE)
#  报告是评价结论的呈现：即使前置步骤都做完了，只要没真正执行过定量评价，
#  也不应凭空推算出结论。
pid3 = new_patch("N")
add_units(pid3); add_constraint(pid3, passed=True);
    add_data(pid3)
with app.app_context():
    pr = wf.report_progress(pid3)
ok(not pr["ready"], "无评价记录 → 报告流程判定为未就绪")
ok(bool(pr["blocking"]) and pr["blocking"][0]["step"] ==
    "evaluate",
    "拦原因指向「执行适宜性评价」")
r = get_report(pid3)
ok(r.status_code == 302 and results_of(pid3) == 0,
    "直接访问 /report/<id> 被退回，且不产生任何评价结果",
    "HTTP %d" % r.status_code)
#  真正执行评价之后，报告才放行，且取值与评价记录一致
r = post_eval(pid3)
ok(results_of(pid3) == 1, "先执行评价 → 产生评价记录")
with app.app_context():
    pr = wf.report_progress(pid3)
    rec = pr["result"]
ok(pr["ready"] and rec is not None, "有评价记录后，报告流程就绪")
rr = get_report(pid3)
body = rr.get_data(as_text=True)
ok(rr.status_code == 200, "此时可打开得分明细页", "HTTP %d" %
    rr.status_code)
ok(("%.4f" % rec.final_score) in body,
    "报告数值取自评价记录，而非重新计算",
    "F=%.4f" % rec.final_score)
cleanup(pid3, "N")
print()
print(LINE)
print("五、内置演示数据的清理不得波及用户自建数据")
print(LINE)
#  回归保护：seed_demo.clean_case() 曾执行 Project.query.delete(),
#  导致用户新建的项目在每次启动时被连带删除。
import seed_demo as sd
with app.app_context():
    keep_p = Project(name="用户自建项目-守护", code="KEEP-001")
```

```
    db.session.add(keep_p); db.session.flush()
    keep_patch = MinePatch(project_id=keep_p.id,
    name="用户自建图斑 - 守护")
    db.session.add(keep_patch); db.session.flush()
    keep_pid = keep_p.id
    keep_patch_id = keep_patch.id
    db.session.commit()
with app.app_context():
    sd.clean_case(reset=False)
with app.app_context():
    still_p = db.session.get(Project, keep_pid)
    still_patch = db.session.get(MinePatch, keep_patch_id)
ok(still_p is not None, "清理演示数据后，用户自建项目仍存在")
ok(still_patch is not None, "清理演示数据后，用户自建图斑仍存在")
# 演示数据清掉后应能重新写入，保证演示环境可恢复
sd.seed()
with app.app_context():
    demo_back = Project.query.filter_by(code=sd.DEMO_CODE).f
    irst()
ok(demo_back is not None, "演示案例可重新写入（幂等恢复）")
with app.app_context():
    m = db.session.get(MinePatch, keep_patch_id)
    if m:
        db.session.delete(m)
    p = db.session.get(Project, keep_pid)
    if p:
        db.session.delete(p)
    db.session.commit()
print()
print(LINE)
print("六、同一图斑不得重复划分同类型微地质单元")
print(LINE)
# 论文 4.1 节把图斑分为平台、碎石边坡、岩质边坡三类，每类至多一个。
# 允许重复会让面积占比、按类型组织的指标数据与批量导入对不上号。
pid4 = new_patch("D")
c.post("/unit/%d/add" % pid4, data={"unit_type": "平台",
    "area": "1.0"},
       follow_redirects=True)
with app.app_context():
    n1 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n1 == 1, "首次添加「平台」成功", "单元数 %d" % n1)
r = c.post("/unit/%d/add" % pid4, data={"unit_type": "平台",
    "area": "2.0"},
            follow_redirects=True)
with app.app_context():
    n2 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n2 == 1, "重复添加同类型被拒绝", "单元数 %d" % n2)
ok("同一类型只能划分一个" in r.get_data(as_text=True), "给出重复提示")
r = c.post("/unit/%d/add" % pid4, data={"unit_type": "火山口",
    "area": "1.0"},
```

```
            follow_redirects=True)
with app.app_context():
    n3 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n3 == 1, "非法单元类型被拒绝", "单元数 %d" % n3)
c.post("/unit/%d/add" % pid4, data={"unit_type": "碎石边坡",
    "area": "3.0"},
        follow_redirects=True)
with app.app_context():
    uid = MicroUnit.query.filter_by(patch_id=pid4,
                                    unit_type="碎石边坡").first(
    ).id
r = c.post("/unit/edit/%d" % uid,
            data={"unit_type": "平台", "area": "3.0"},
    follow_redirects=True)
with app.app_context():
    types = sorted(u.unit_type for u in
                   MicroUnit.query.filter_by(patch_id=pid4)
    all())
ok(types == ["平台", "碎石边坡"], "改类型时同样拦截重复", "、".join(types))
cleanup(pid4, "D")
print()
print(LINE)
if _fails:
    print("存在未通过项：")
    for f in _fails:
        print("  -", f)
    sys.exit(1)
print("流程门禁自检全部通过 √")
print(LINE)
#  ==== 测试/test_weights.py ====
#  -*- coding: utf-8 -*-
"""权重生效链路自检：AHP 算出的权重必须真正参与评价计算。
此前 weight_schemes / weight_items 两张表从未被写入，评价引擎也只用
代码里的论文表 3 默认权重，相当于权重模块算完即弃。本测试守住这条链路：
    默认 → 论文表 3
    保存 AHP 权重 → 评价改用新权重，结果随之变化
    恢复默认 → 结果回到论文取值
"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
#  使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil
    e__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
from app.models import (db, Project, MinePatch, MicroUnit,
    IndicatorData,
```

```
                                ConstraintCheck, EvaluationResult,
    WeightScheme,
                                WeightItem)
from app.services import workflow as wf
from app.services import weight_service as ws
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
_fails = []
def ok(flag, label, extra=""):
    print("  %s %-46s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    if not flag:
        _fails.append(label)
TAG = "W"
UNIT_TYPE = "碎石边坡"
def build_case():
    """建一个只含单一微地质单元的图斑，便于观察权重变化对得分的影响。"""
    with app.app_context():
        p = Project(name="权重测试-%s" % TAG, code="WEIGHT-%s"
    % TAG)
        db.session.add(p); db.session.flush()
        patch = MinePatch(project_id=p.id, name="权重测试图斑-%s"
    % TAG)
        db.session.add(patch); db.session.flush()
        db.session.add(MicroUnit(patch_id=patch.id,
    unit_type=UNIT_TYPE,
                                 area=1.0, ratio=1.0))
        db.session.flush()
        u = MicroUnit.query.filter_by(patch_id=patch.id).fi
    st()
        db.session.add(IndicatorData(
            unit_id=u.id, soil_ratio=0.34, thickness=10,
    ph=9.24,
            organic_matter=4.89, coverage=87.92,
    shannon=0.60,
            slope=28.75, aspect="阳坡"))
        db.session.add(ConstraintCheck(
            patch_id=patch.id, policy_pass=True,
    location_pass=True,
```

```
                pollution_pass=True, geo_pass=True,
        farmland_pass=True,
                passed=True, reason="全部通过"))
            db.session.commit()
            return patch.id
def cleanup(pid):
        with app.app_context():
            from app.models import ScoreDetail
            rids = [r.id for r in EvaluationResult.query.filter_
        by(patch_id=pid)]
            if rids:
                ScoreDetail.query.filter(ScoreDetail.result_id.i
        n_(rids)).delete(
                    synchronize_session=False)
            EvaluationResult.query.filter_by(patch_id=pid).delet
        e()
            ConstraintCheck.query.filter_by(patch_id=pid).delete
        ()
            uids = [u.id for u in MicroUnit.query.filter_by(pat
        h_id=pid)]
            if uids:
                IndicatorData.query.filter(IndicatorData.unit_id
        .in_(uids)).delete(
                    synchronize_session=False)
            MicroUnit.query.filter_by(patch_id=pid).delete()
            m = db.session.get(MinePatch, pid)
            if m:
                db.session.delete(m)
            p = Project.query.filter_by(code="WEIGHT-%s" %
        TAG).first()
            if p:
                db.session.delete(p)
            db.session.commit()
def run_eval(pid):
        c.post("/evaluate/%d" % pid, data={},
        follow_redirects=False)
        with app.app_context():
            return EvaluationResult.query.filter_by(patch_id=pid
        ) \
                .order_by(EvaluationResult.id.desc()).first().fi
        nal_score
print(LINE)
print("一、默认应使用论文表 3 的权重")
print(LINE)
with app.app_context():
        ws.reset_weights(UNIT_TYPE)
        w, label, scheme = ws.effective_weights(UNIT_TYPE)
ok(scheme is None and "论文表" in label, "未配置时权重来源为论文表 3",
        label)
ok(abs(w["l1"]["土壤指标"] - 0.44) < 1e-9, "一级权重取自论文表 3",
```

```
    "土壤指标 %.2f" % w["l1"]["土壤指标"])                                    12203
pid = build_case()                                                          12204
base_score = run_eval(pid)                                                  12205
print("  默认权重下的评价值 F = %.4f" % base_score)                            12206
print()                                                                     12207
print(LINE)                                                                 12208
print("二、保存 AHP 算出的权重后，评价应改用新权重")                              12209
print(LINE)                                                                 12210
with app.app_context():                                                     12211
    # 模拟 AHP 计算结果：把土壤指标权重提高、植被与地形降低（仍和为 1）              12212
    ws.save_weights(UNIT_TYPE, [("soil", 0.70), ("veg",                     12213
    0.20),                                                                  12214
                                ("terrain", 0.10)])                         12215
    n_items = WeightItem.query.count()                                      12216
ok(n_items >= 3, "weight_items 表已写入权重条目", "%d 条" % n_items)          12217
with app.app_context():                                                     12218
    w2, label2, scheme2 = ws.effective_weights(UNIT_TYPE)                   12219
ok(scheme2 is not None and "自定义" in label2, "权重来源变为自定义方案",        12220
    label2)                                                                 12221
ok(abs(w2["l1"]["土壤指标"] - 0.70) < 1e-9, "新权重已生效",                   12222
    "土壤指标 %.2f" % w2["l1"]["土壤指标"])                                   12223
ok(abs(w2["l1"]["植被指标"] - 0.20) < 1e-9, "同批保存的其他项也已生效")         12224
ok(abs(w2["l2"]["土壤指标"]["土石比"] - 0.25) < 1e-9,                         12225
    "未覆盖的二级权重仍保留论文取值",                                          12226
    "土石比 %.2f" % w2["l2"]["土壤指标"]["土石比"])                           12227
new_score = run_eval(pid)                                                   12228
print("  新权重下的评价值 F = %.4f" % new_score)                               12229
ok(abs(new_score - base_score) > 1e-4,                                      12230
    "权重变化确实改变了评价结果",                                              12231
    "%.4f → %.4f" % (base_score, new_score))                                12232
print()                                                                     12233
print(LINE)                                                                 12234
print("三、结果应记录所用权重来源")                                            12235
print(LINE)                                                                 12236
import json                                                                 12237
with app.app_context():                                                     12238
    rec = EvaluationResult.query.filter_by(patch_id=pid) \                  12239
          .order_by(EvaluationResult.id.desc()).first()                     12240
    stored = json.loads(rec.unit_scores)                                    12241
ok(any(u.get("weight_source") for u in stored),                             12242
    "评价记录中标注了权重来源",                                               12243
    stored[0].get("weight_source") or "")                                   12244
print()                                                                     12245
print(LINE)                                                                 12246
print("四、恢复默认后应回到论文取值")                                          12247
print(LINE)                                                                 12248
with app.app_context():                                                     12249
    ws.reset_weights(UNIT_TYPE)                                             12250
    w3, label3, scheme3 = ws.effective_weights(UNIT_TYPE)                   12251
ok(scheme3 is None and "论文表" in label3, "已恢复为论文表 3 权重",           12252
```

```
    label3)                                                          12253
back_score = run_eval(pid)                                           12254
print("  恢复后评价值 F = %.4f（默认时 %.4f）" % (back_score,          12255
    base_score))                                                     12256
ok(abs(back_score - base_score) < 1e-9, "评价结果与默认时一致")       12257
print()                                                              12258
print(LINE)                                                          12259
print("五、权重首页概览")                                            12260
print(LINE)                                                          12261
with app.app_context():                                              12262
    rows = ws.summary()                                              12263
ok(len(rows) == 3, "三种微地质单元均有权重来源说明", "%d 行" % len(rows))  12264
r = c.get("/ahp/")                                                   12265
body = r.get_data(as_text=True)                                      12266
ok(r.status_code == 200 and "当前生效权重" in body, "权重首页正常展示")  12267
print()                                                              12268
print(LINE)                                                          12269
print("六、一致性不通过的矩阵不得保存生效")                          12270
print(LINE)                                                          12271
with app.app_context():                                              12272
    ws.reset_weights(UNIT_TYPE)                                      12273
n = 3                                                                12274
# 明显矛盾的矩阵：a12=9, a13=1/9, a23=9 —— CR 必然超限               12275
mats = [(0, 1, "9"), (0, 2, "0.1111"), (1, 2, "9")]                  12276
data = {"csrf_token": "", "action": "save"}                          12277
for i, j, v in mats:                                                 12278
    data["m_%d_%d" % (i, j)] = v                                     12279
r = c.post("/ahp/matrix?unit_type=%s&level=1" % UNIT_TYPE,           12280
    data=data,                                                       12281
               follow_redirects=True)                                12282
with app.app_context():                                              12283
    _, label4, scheme4 = ws.effective_weights(UNIT_TYPE)             12284
ok(scheme4 is None, "CR 超限的矩阵未被保存生效", label4)             12285
ok("一致性" in r.get_data(as_text=True), "页面给出了未通过一致性的提示")  12286
cleanup(pid)                                                         12287
print()                                                              12288
print(LINE)                                                          12289
if _fails:                                                           12290
    print("存在未通过项：")                                          12291
    for f in _fails:                                                 12292
        print("  -", f)                                             12293
    sys.exit(1)                                                      12294
print("权重生效链路自检全部通过 √")                                  12295
print(LINE)                                                          12296
# ==== app/services/__init__.py ====                                 12297
# -*- coding: utf-8 -*-                                              12298
# ==== app/utils/__init__.py ====                                    12299
# -*- coding: utf-8 -*-                                              12300
# ==== app/views/__init__.py ====                                    12301
# -*- coding: utf-8 -*-                                              12302
```